



INDIAN EXPRESS UPSC IAS EDITION HD 20~07~2026
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All the topics of this UPSC IAS Edition are directly or
indirectly important for the prelims & main examination.
There are some topics which can be coded in answer writing
of other topics in the main exam.

Govt set to bring Bill to criminalise insult to Vande Mataram

Jatin Anand
New Delhi, July 19

THE GOVERNMENT is all set to bring a Bill in the Rajya Sabha on the first day of the Monsoon Session that will make any obstruction or insult to the national song “Vande Mataram” a criminal offence—at par with the national anthem “Jana

Gana Mana”.
According to the List of Business for the Rajya Sabha for Monday, “The Prevention of Insults to National Honour (Amendment) Bill, 2026” is likely to be introduced by Union Home Minister Amit Shah.
The Bill proposes to amend the Prevention of Insults to
»CONTINUED ON PAGE 2



PV Sindhu with coach Irwansyah after winning the Japan Open 2026, in Tokyo, Sunday. BWF BADMINTONPHOTO

Sindhu answers doubters with Japan Open crown: ‘Had tears in my eyes’

Shivani Naik
Mumbai, July 19

TOKYO HAS always given P V Sindhu a second wind.
The 31-year-old had not won a Super 750 grade title — the second highest tier after Super 1000s — since her World Championship win in 2019. And the murmurs had grown louder, casually nudging her to retire as

she had achieved plenty.
On Sunday, Sindhu shut out the ear-shattering cacophony of Japanese fans cheering on the home favourite — triple World Champion and World No 3 Akane Yamaguchi — to finally claim the Japan Open with a scorching 21-17, 21-17 win. The World No. 10 also turned down the decibel levels of those
»CONTINUED ON PAGE 2

INTERIM DEAL CRUMBLES; HORMUZ TRAFFIC LARGELY STALLED

US strikes to ‘punish’ Iran after 2 of its soldiers killed; Tehran intensifies fire on Gulf states

Missiles launched at Jordan, risk of spillover into Israeli territory, warns Tel Aviv

Jon Gambrell & Sam Metz
Dubai, July 19

THE UNITED States launched more airstrikes at Iran on Sunday in response to the killing of US troops, and Iran fired missiles toward Jordan that risked widening the conflict into neighbouring Israel.
Step by step, the US and Iran have returned closer to all-out war as last month’s interim deal — meant to permanently end the fighting — has crumbled and shipping traffic in the Strait of Hormuz has largely stalled.
The US military said the la-



Kuwaiti PM Sheikh Ahmad Abdullah Al-Ahmad Al-Sabah visits a person injured in the Iranian attack, in Kuwait City, Sunday. REUTERS

test strikes targeted Iran’s paramilitary Revolutionary Guard — a retaliation for Friday’s killing of troops in Jordan. The campaign, now in its second week, has seen the US target bridges, water desalination plants and electrical facilities in Iran. Tehran has hit US-allied countries throughout West Asia. Kuwait, Jordan and Bahrain

again activated air defences for incoming Iranian drones and missiles. Israel warned that missiles launched toward neighbouring Jordan could cause fire to spill over into Israeli territory for the first time in weeks.
The US Central Command said it hit “Iranian military coastal surveillance and air
»CONTINUED ON PAGE 2

• SHOWPIECE OF MUSEUM IN RS 4,000-CRORE MARITIME COMPLEX NEAR LOTHAL

Coming soon in Gujarat: A life-size Harappan town with markets, dockyards

Divya A
New Delhi, July 19

VERY SOON, visitors to Lothal, one of the most prominent outposts of the Indus Valley Civilisation, will be able to get a feel of walking through Harappan streets, dockyards, markets and civic spaces.

Just a few minutes away from the archaeological site, about 80 km from Ahmedabad, architects are recreating parts of the historic town for a museum that will be part of the Rs 4,000-crore National Maritime Heritage Complex

(NMHC), whose first phase will be inaugurated within weeks.
Lothal is believed to have existed around 4,500 years ago, with excavations at the site revealing the world’s oldest known artificial dock, connected to an old course of the Sabarmati river.
According to officials, the first phase of the complex cost an estimated Rs 775 crore, and includes six museum galleries, a jetty walkway, and a display of maritime artefacts spanning ancient, medieval, colonial and modern times.
But the centrepiece of the

375-acre project, cleared by the Union Cabinet in 2024, remains the recreation of Lothal town.
“Rather than displaying history behind glass, we wanted people to physically walk through reconstructed Harappan streets, dockyards, markets, and civic spaces to understand how advanced these settlements were,” Karl Wadia, Principal Architect at Architect Hafeez Contractor, who is heading the project, told *The Indian Express*.
The firm was signed on to design the project by the Union Ministry of Ports and Shipping, the nodal ministry involved, with the total project cost



Graphic rendering of the upcoming National Maritime Heritage Complex in Lothal. ARCHITECT HAFEEZ CONTRACTOR

pegged at around Rs 4,000 crore, officials said. “The idea

was to create a world-class cultural and educational destina-

tion where visitors can experience India’s 4,500-year maritime history in an immersive, engaging way,” Wadia said.
The complex brings together museums, public landscapes, historic recreations, research facilities, eco-tourism and entertainment programming within a single integrated complex. “Alongside maritime artefacts from ancient, medieval, colonial, and modern periods, we are incorporating large-scale audio-visual projections, tactile exhibits, underwater-themed galleries, suspended ship installations, and digital storytelling environments that trace the

evolution of Indian seafaring traditions,” Wadia said.
Beyond the maritime museum, the complex will include a lighthouse museum, a 5D theatre, children’s galleries, waterfront promenades, public plazas, eco-resorts and themed attractions. “The maritime museum is really the heart of the entire NMHC project. At roughly 7 lakh sq ft, it will be the world’s largest maritime museum. The design is inspired by the archaeological remains of the ancient city of Lothal,” Wadia said.
The museum comprises monumental stone-clad forms raised on elevated plinths, rein-

terpreting the flood-resistant citadels of Harappan settlements. Two inclined forms at the top resemble ship hulls, inspired by the righting-lever principle used in naval engineering, according to the concept.
Inside, visitors will move through 14 interconnected galleries organised around a recreated Lothal settlement. The experience begins in an 18-metre-high arrival hall and unfolds through galleries arranged around a circulation spine, featuring large water tanks with staged displays, immersive projections, suspended
»CONTINUED ON PAGE 2



Expanding Ties

External Affairs Minister S Jaishankar with Zanzibar President Hussein Ali Mwinyi in New Delhi, Sunday. The two sides discussed expanding cooperation in higher education, capacity building, water supply, health, AI, digital and other priority sectors. PTI

BHANIYAWALA-RISHIKESH ROAD

Plan to cut 4,000 trees on hold, took note of concerns, says Dhami

Aiswarya Raj
Dehradun, July 19

FOLLOWING PROTESTS and an FIR against dozens, the Uttarakhand government has suspended felling of trees for the Bhaniyawala-Rishikesh road in Dehradun.

The National Highways Authority of India project proposes to widen the 19-km stretch near the airport and had earmarked 4,369 trees for felling.

Chief Minister Pushkar Singh Dhami said he has directed the Chief Secretary and officials to hold detailed consultations with all stakeholders, residents, public representatives, and experts before the resumption of tree-felling.

The NHAI had claimed that it deposited more than Rs 1.97 crore towards compensatory afforestation and its maintenance for the next ten years. And to facilitate compensatory afforestation, 40 hectares of non-forest land have been transferred by the State Government to the Forest Department for the development of new forests.

However, the episode has caused anger among the locals as the road passes through the Shivalik elephant reserve. People tried to disrupt the felling by mobilising in the area. On July 12, an FIR was registered against unknown individuals for allegedly “using criminal force to deter public servants from discharging their duty”.

As the momentum against the road-widening project increased, CM Dhami on Saturday announced that the tree felling will be suspended until a consensus is reached with all the stakeholders. “Over the past

few days, I have taken serious note of the concerns and suggestions expressed by numerous citizens, environmentalists, and residents regarding the Dehradun–Rishikesh Four/Six-Lane Project. This project is an important infrastructure initiative of the National Highways Authority of India (NHAI). Work on the project was being carried out in compliance with the directions of the Hon’ble High Court and after following all required statutory and environmental approvals and procedures,” he said.

Keeping wildlife conservation in mind, the project also provides for a 3.5-kilometre-long elephant underpass and special culverts for the movement of smaller wildlife, he claimed. “Development is essential for us, but no decision will be taken by ignoring public sentiment, the environment, or local interests,” the CM said.

A contempt petition filed before the High Court regarding a stay on the tree felling was dismissed.

Environmentalists say the road widening project will impact the eighth, ninth and tenth corridors in the Shivalik Elephant reserve and parts of the elephant corridors.

In 2020, amid talks of airport expansion, the Ministry of Environment, Forest and Climate Change (MoEF) had asked the Uttarakhand government to consider avoiding sensitive areas of the Shivalik Elephant Reserve while exploring land suitable to use for the expansion of Dehradun’s Jolly Grant Airport. The government had attempted to denotify the Shivalik Elephant reserve, which was stayed by the High Court.

Lothal

artefacts and underwater-themed installations.

The narrative spans ancient Harappan trade networks, Indian Ocean commerce, South Indian naval power, colonial encounters and modern Indian shipping, said design firm officials. At the centre of the museum will be an 80m x 80m living-history recreation of Lothal town. “We wanted visitors to engage with history spatially and experientially rather than simply viewing it as an exhibit,” the officials said.

Asked how the project engages with the legacy of the Indus Valley Civilisation, Wadia said Lothal’s history as a significant maritime site made it a natural choice as the project’s grounding site, being home to what is considered the world’s earliest known man-made dockyard. “Historically, Lothal connected the Indian subcontinent to Mesopotamia, Egypt, and West Asia through maritime trade, so origins of Indian seafaring culture are deeply rooted there. We wanted the project to reconnect contemporary audiences with the sophistication, innovation and global outlook of Indus Valley Civilisation,” he said.

Footage released by the US military appeared to show strikes carried out by fighter jets and by Tomahawk cruise missiles launched from the sea. One target site appeared to be in a valley of a mountainous region.

Iran’s atomic energy agency said US strikes targeted the construction site of a planned nuclear power plant in the southwest, state television reported.

US strikes

defense facilities, maritime capabilities and missile and drone storage sites.” It said the attack was designed to degrade Iran’s ability to control the strait and “swiftly punish Islamic Revolutionary Guard Corps forces.”

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Iran’s atomic energy agency said US strikes targeted the construction site of a planned nuclear power plant in the southwest, state television reported.

The International Atomic Energy Agency said the site, in the “very early stages of construction,” contained no nuclear material.

Jordan’s military said it shot down several Iranian missiles. The country hosts major US bases and relies on US air-defence systems. The missiles did not cause casualties or damage, according to Jordan’s military.

Israel’s military said Iranian missiles launched toward the Jordanian port city of Aqaba just across the border could spill over, warning Israelis to expect the first air raid sirens in weeks. Eilat, the Israeli city that neighbors Aqaba, cited security officials as saying two interceptors were launched from its outskirts to prevent the fall of debris. “We

are prepared to immediately resume combat,” the Israeli military chief of staff said.

Kuwait said one of its power and water desalination plants was attacked for the second time in two days, causing fires.

US President Donald Trump has threatened to target Iran’s power stations and bridges to try to compel Tehran to loosen its hold on the Strait of Hormuz.

(Reuters adds: In a written statement carried by Iranian state media late on Saturday, Supreme Leader Ayatollah Mojtaba Khamenei said US actions had shown that Trump’s signature was “utterly worthless and devoid of credibility.” The statement warned of “even heavier costs and further humiliation” for the US.) AP

Sindhu

Indians who had gracelessly asked her to leave her stomping ground. Sindhu’s statement win said she’s going nowhere.

After beating her certified nemesis, Sindhu fell to the floor, and shed tears behind her cupped face. “I had tears in my eyes because it was important for me to win,” she said after the title triumph. “I was really really focusing hard and working hard on myself. I kept believing I could do it. Lot of people were like, what’s happening? Is it (career) done? Or whatever it is. But still, I believed in myself and my family and my coaches supported me.”

For some time now, the top step of the podium had seemed out of reach for Sindhu. At the 2021 Tokyo Olympics, after being thrashed in the semifinals by Tai Tzu-ying, she had regrouped to beat He Bingjiao and claim bronze. But Sindhu hadn’t beaten Yamaguchi in a completed match since May 2022. Her last title had been a lowly Super 300, 19 months ago. In fact, this was only her third Super750 plus title of her career, given she has more Olympic and World medals than big triumphs on the circuit.

Sindhu stressed the importance of winning a final. “Playing in the final is one thing. But to win, stand on the podium, win that gold is a different thing altogether,” she said.

Against Yamaguchi, Sindhu deployed all her power, strength and speed as well as defensive stubbornness. Yamaguchi was heart-broken after the loss. “I am so so upset, I tried to make points from my defence, but her attack

was very good. She was so aggressive, the smash so strong...” the Japanese player said, walking away dejected as the home crowd went silent.

It took Indonesian coach Irwansyah to drag Sindhu out of the defensive zone she had got into after the 2019 world title. Six coaches in the last seven years had come and gone, unable to rewind her attacking instinct completely.

While accepting wishes from celebrities and politicians, Sindhu did not forget to thank her husband Venkata Datta Sai, a sports geek, data specialist and first cheerleader who, she tweeted, had taken on the onerous task of “rebuilding her team”.

Coach Irwansyah made tactical changes — smoothing her movement, sharpening her defence-to-offence transitions, using the net for follow-up kills, activating the drop shots to ease over-reliance on smashes — to give her confidence.

Sindhu also brought in fitness maestro Wayne Lombard, a South African who watched the final on a stream, from his hospital bed while recovering from surgery. The biokinetics PhD holder has worked with India’s boxing and hockey teams. For Sindhu, he designed programmes to primarily increase her strength.

Sindhu also shared that she had worked on staying calm — probably the main reason why previous finals had been fluffed when she got agitated. “Sometimes when leading you want to finish rallies quicker, you tend to make simple unforced errors. But the coach said, ‘It’s OK, focus on the next point’. I let go of what happened, and focused on the next point. That made a difference,” she said.

Vande Mataram

National Honour Act, 1971, which makes desecration of or disrespect toward the country’s national symbols, including the national flag, the Constitution and the national anthem, a criminal offence, with a punishment of up to three years’ imprisonment as well as provisions for a fine.

The proposed move drew a sharp reaction from the CPM, with Rajya Sabha MP John Brittas calling for the Bill’s withdrawal,

stating in a letter to Shah that it “risks unsettling a delicate constitutional balance founded upon pluralism, liberty of conscience and voluntary patriotism”.

According to a copy of the proposed legislation circulated among MPs, the Bill seeks to substitute the existing Section 3 of the 1971 Act to effectively bring “Vande Mataram” at par with “Jana Gana Mana” with regard to alleged acts of preven-

tion or disturbance against singing them, with the same penal provisions — three years, a fine, or both.

“Whoever, intentionally — (a) prevents the singing of the National Anthem or the National Song; or (b) causes disturbance to any assembly engaged in such singing, shall be punishable with imprisonment for a term which may extend to three years, or with fine, or with both,” the Bill states.

Last year, a special discussion was held in both Houses in December to mark the 150th anniversary of the na-

tional song, with sittings taking place over four days during the Winter Session.

During the debate, the BJP-led NDA Government locked horns with the Opposition, mainly the Congress. PM Narendra Modi accused the Congress of having truncated the song to appease the Muslim League and Mohammad Ali Jinnah, adding that omitting important stanzas had put the country on the path of “appeasement politics” that ultimately led to Partition.

In response, Congress leader and Wayanad Lok Sabha

MP Priyanka Gandhi Vadra invoked the “chronology” behind the song and accused the PM of misrepresenting the facts in the matter.

Reading excerpts of correspondence between Jawaharlal Nehru and Subhash Chandra Bose, Vadra underlined that the first two stanzas “were always sung” and that the ones added later could be interpreted as communal. She asserted that it was following the Congress Working Committee’s resolution on October 28, 1937, that “Vande Mataram” was declared the national song.

According to the amendment Bill’s statement of objectives, there is at present no specific legal provision to prevent insults to the singing of “Vande Mataram”.

“Therefore, in order to prohibit any person from intentionally preventing the singing of the National Song (Vande Mataram) or causing disturbance to any assembly engaged in such singing, it is proposed to amend Section 3 of the said Act to include the National Song also within its ambit, so as to make such acts punishable,” the objectives state.

TRACES 200 YEARS OF THE RIVER’S TRANSFORMATION

Yamuna floodplains shrank by a third in a century: Study

Sophiya Mathew
New Delhi, July 19

YAMUNA’S SHRINKING floodplains and increasing flood vulnerability in Delhi are the result of changes that have unfolded over more than two centuries, according to a study published in April this year in the Journal of the Geological Society of India by researchers from the University of Delhi (DU) and the Indian Institute of Science Education and Research (IISER), Bhopal.

Reconstructing the river’s evolution using historical maps and satellite imagery, the study noted that barrages, embankments, and rapid urbanisation have altered the Yamuna’s natural seasonal flow and its ability to accommodate high flows during heavy rainfall.

The study has estimated that rapid urbanisation has narrowed the river’s channel by about 68%, reduced its long-term channel-maintaining, or formative discharge by nearly 89%, and disconnected around one-third of its floodplains.

Formative discharge is crucial, as it provides the flow a river needs to naturally shape its channel, remove accumulated sediment, and maintain its capacity to safely accommodate heavy rains.

“These changes in river width, discharge, and land use... have reduced the natural capacity of the river to adjust to extreme events,” the authors wrote. The study, authored by researchers Tanya, Sampat Kumar Tandon, Kumar Gaurav, and Vimal Singh, analysed a 50-km stretch of the Yamuna flowing through Delhi using historical maps dating back to 1799, Survey of India records, and satellite imagery up to 2024. By tracing changes in the



The study shows that the river’s channel has narrowed by 68% over the last two centuries. FILE

river’s width over time, the researchers reconstructed its hydrogeomorphic evolution, meaning how evolving landforms dictate water flow, and estimated changes in its formative discharge. In doing so, they studied the long-term average flow that shapes a river channel over decades.

The construction of the Tajewala, Wazirabad, and Okhla barrages has been a key driver of the Yamuna’s altered flow regime, according to the study. It noted that nearly 90% of the Yamuna’s flow is diverted at the Hathnikund Barrage upstream, leaving the Delhi stretch of the river with very little freshwater during the lean season. The study called the Delhi stretch “an ideal example of an anthropogenically modified river reach.”

Comparing floodplain maps from 1912 with present-day data, the researchers estimated that around 45 sq km, roughly one-third of the river’s floodplains in Delhi, have been disconnected by embankments over the last century.

These disconnected flood-

plains, the study said, have since seen agriculture, settlements, and infrastructure development, reducing the river’s natural space to spread during floods.

The study also documented a sharp decline in sediment deposits that form within the river. Their total area has shrunk from around 20 sq km in 1985 to about 4 sq km in 2020, with much of it converted for agriculture and other uses.

The implications became evident during the July 2023 floods. Although the flood carried less water than the 1978 flood, the Yamuna reached a higher water level in Delhi. According to the study, embankments and urban development have progressively constricted the river corridor, preventing floodwaters from spreading laterally across floodplains and forcing them into a narrower channel instead. “The consequences of these developments have reduced the natural buffer capacity of the Yamuna River corridor... thereby increasing the vulnerability to flooding events,” the paper stated.

“The river requires a channel-maintenance discharge,” said Vimal Singh, faculty member at DU’s Department of Geology and one of the study’s authors. He stressed that the study estimates the river’s long-term flow responsible for maintaining the channel’s shape, not the river’s discharge in recent years. Singh said that ensuring environmental flows remains essential for sustaining aquatic ecosystems, but added that restoring the Yamuna can no longer be viewed solely through the lens of river flow. “River ecology is now much more complex because livelihoods are dependent on the river,” he said.

He added that upgrading sewage treatment plants, increasing the use of treated wastewater instead of potable water wherever feasible, improving the condition of major drains such as the Najafgarh (Sahibi) drain, and reducing the growing dependence of Delhi’s expanding population on the Yamuna for freshwater would all be important measures for improving the river’s health.

The authors recommended that a “long-term hydrogeomorphic perspective” is necessary to understand how the river transitioned into its current “degraded situation.”

While the study does not make specific policy recommendations, it cautioned that further constricting the river through embankments and continued encroachment on floodplains could increase flood vulnerability by reducing the Yamuna’s natural buffer capacity.

It also called for floodplains to be recognised as dynamic systems rather than spaces to be engineered or reclaimed for development.

Jail for 'illegal' live-in: MP Cabinet clears UCC draft Bill

Set to be introduced in Assembly during Monsoon Session

Anand Mohan J
Bhopal, July 19

THE MADHYA Pradesh Cabinet on Sunday approved the draft Uniform Civil Code (UCC) Bill, proposing a common civil law for all residents that bans polygamy, criminalises practices such as triple talaq and provides for potential imprisonment of couples in live-in relationships if they don't declare their status.

The draft legislation is expected to be introduced in the Assembly during the Monsoon Session beginning Monday.

According to the Bill, couples in a live-in relationship will be legally required to file a "declaration of live-in relationship" with the registrar within one month of starting to live together. If a partner is under 21, information about the "start and end of the live-in relationship will be sent to their parents/guardians". The registrar will also send this record to the local police station.

The penalties proposed in the Bill include three months' imprisonment or a fine of Rs 10,000 for not registering the declaration within a month. Giving false information can attract three months' imprisonment and a fine of Rs 25,000, and failing to file the declaration even after a registrar's notice can attract up to 6 months' imprisonment and a fine of Rs 25,000.

The Bill has provisions to

consider children born of a live-in relationship to have full inheritance rights. If the male partner in the relationship deserts the woman, she "can claim maintenance through a competent court just as a legal wife would," as per the Bill.

The code makes monogamy the norm across all communities. A person may be married to only one living spouse at a time as per the Bill. The minimum age for marriage is fixed at 21 years for men and 18 years for women. Marriages involving invalid consent and prohibited relationships (unless permitted by custom) are barred.

Oral divorce or decisions by informal panchayats are "declared entirely illegal". A marriage can now be dissolved only on the clear, statutory grounds specified in law. "Accepting, promoting, or forcing degrading or humiliating conditions such as 'nikah halala' for remarriage to the same spouse after divorce will be treated as a punishable criminal offence," the Bill stated.

Keeping in mind the sizeable population of ST, the Bill proposes that the law would not apply to STs covered under Articles 342 and 366(25) of the Constitution (such as Bhil, Gond, Korku, Baiga, Sahariya and Bharia).

The Bill also proposes to ban the word illegitimate and give every child equal legal status. Sons and daughters are given equal rights to inherit property, regardless of marital status. Widows and widowers will also be treated equally in the deceased's property.

Yadav had formed a committee headed by retired Supreme Court judge Justice Ranjana Desai to prepare the draft.

J&K: 18 die as flash floods, landslides wreak havoc

Arun Sharma
Jammu, July 19

AT LEAST 18 PEOPLE, including several children, were killed, a large number of vehicles were swept away, and several residential and commercial buildings were damaged after overnight heavy rain led to flash floods and landslides in the early hours of Sunday in Jammu and Kashmir's border districts of Poonch and Rajouri.

Sixteen people died in Poonch district alone, and Rajouri town saw large-scale property damage.

In Poonch, 15 of the 16 deaths took place in the Surankote area — eight in Murrah, four in Sangla, and one each at Sanglani, Marhote and Dogrian Baffliaz. Nazia Kousar (28) died in the Noon Bandi area near Poonch when her house collapsed. Her husband, Mohammad Hafiz, and three children, aged between two and six, were rescued from under the debris.

Around 50 private vehicles parked at the Bela Colony bus stand area of Rajouri were washed away in a flash flood. According to locals, the bus stand, made operational in November 2024, was built on a mound in a river that had been dry for a few decades.

However, the river was flooded on Sunday by water brought into it from the nullahs coming from the Darhal and Thanamandi sides. The bus stand subsequently got sub-



In Mendhar area of Poonch district on Sunday. PTI

merged under flood waters, and the vehicles parked there during the night were washed away.

The flood waters also entered nearby slums, and a woman named Pinki Devi was reported to have been swept away, officials said.

A local government official said the death toll was not high in the area because there was no one present at the bus stand area when the flooding occurred.

J&K Chief Minister Omar Abdullah, who had gone to New Delhi to sit on dhama at Jantar Mantar on Monday demanding the restoration of statehood for Jammu and Kashmir, said he would return to Jammu in view of the disaster.

Expressing grief over the loss of lives in Poonch and Ra-

jouri, he said the SDRF, police and all other agencies concerned were engaged in relief and rescue operations, adding that the government will do everything possible to aid and assist affected people who have suffered property loss/damage due to the rains and flash floods.

Lt Governor Manoj Sinha said affected families have already been moved to safety, all departments are on high alert, and that they are prioritising immediate relief and repair work. He urged residents to stay calm and follow official updates. In view of the inclement weather conditions, while various link roads stand blocked due to landslides in the twin border districts, the pilgrimage to Shri Mata Vaishno Devi and Shiv Khori shrines in Reasi district was temporarily sus-

pending.

Meanwhile, authorities in the Jammu division have sounded the alert, following a sharp rise in the water level of the Chenab River and several other streams.

Officials said that all the gates of the Salal Hydroelectric Project in Reasi remained open for the second consecutive day as part of the reservoir flushing process, leading to a significant increase in the discharge and flow of the Chenab River.

In neighbouring Ramban district, police issued a public advisory after the opening of the gates of the Baglihar Hydroelectric Project also caused the river's water level to rise.

Police teams have also been deployed in villages along the Chenab River in Reasi, using public address systems to warn residents against venturing near the river and advising them to keep children and livestock away from its banks.

The district administration has cautioned that the water level in the Chenab may rise suddenly due to continued discharge from the dams and appealed to people to avoid unnecessary travel and stay away from rivers, streams and other water bodies.

According to IMD officials, Reasi received 106.5 mm of rain during the last 24 hours, followed by Rajouri (103 mm), Udhampur (101.6 mm), Katra (85 mm), Poonch (77 mm), Batote (57.2 mm) and Ramban (9 mm) in the Jammu region.

50 spotted deer translocated from Sasan Gir to Mandvi forest

Express News Service
Ahmedabad, July 19

IN A BID TO mitigate incidents of human-wildlife conflict, the Gujarat Forest department has translocated 50 spotted deer from the Sasan Gir Wildlife Division to the Mandvi Forest Range in Surat district.

The initiative is intended to strengthen the natural prey base for carnivores, enhance biodiversity, and support long-term ecological balance in South Gujarat, an official release said on Sunday.

"The translocation was carried out in three phases between May and June this year. A total of 21 spotted deer were released on May 23, followed by 16 on June 18 and another 13 on June 24, taking the total number to 50," the release said.

Conservator of Forests (Surat Circle) Puneet Nayyar said forest officials had carried out habitat assessments to identify sites capable of sustaining the released animals.

"To ensure year-round water availability, the department has developed 10 permanent water points across the release area, providing reliable drinking water for the deer," Nayyar said.



MONDAY, JULY 20, 2026

An uncertain world needs food, fuel, forex buffers

THE ONGOING West Asia conflict has been considered the largest oil shock in history, taking off a fifth of the world's supply that used to transit the Strait of Hormuz daily. Yet, all through this, global crude prices haven't surged like they did in the previous shocks of 2022 or 2008. One reason has been the International Energy Agency-coordinated action, on March 11, for the release of some 426 million barrels by member countries from their emergency oil stockpiles over a four-month period. This quantity made available to the market has put a lid on crude prices, which settled within \$90-110 per barrel in April-May and eased to \$70-80 by mid-June, when the war was seemingly over. It's another thing that the hostilities have resumed, even as the oil buffers are running thin due to the earlier inventory drawdowns.

The same buffer story may play out in food, where the Food Corporation of India's rice and wheat stocks of 121.7 million tonnes (mt) as on June 1 are nearly thrice the required minimum level. Government agencies are also holding over 4 mt of pulses stocks. And it's not only India. The world, too, is awash with ample supplies from the record 2025-26 harvests of wheat, rice, corn, sugar and even soyabean, palm oil and rapeseed. Drawing down of these stocks should act as a cushion against any El Niño-induced production disruptions in the current agricultural year. The contrast with 2022 is, again, instructive. At the start of the Russia-Ukraine war, the world was still recovering from the supply chain dislocations caused by the Covid lockdowns. The buffers then weren't adequate for the global oil and food commodity markets to absorb that war shock. For India, the situation was compounded by the strong El Niño event of 2023-24 that unleashed a prolonged episode of food inflation.

The short point is that accumulating large reserves — whether of food, fuel or money — has become central to the policymaker's arsenal in today's world of rising climate and geopolitical risks. There is a cost to such stockpiling, similar to spending on defence and maintaining a standing army. Their utility is proved only during supply shocks from weather, war or a balance of payments crisis. How much of this cost the government should bear, and the feasibility of stocking various commodities, is both a fiscal and strategic calculation. All the more reason for the government to have an integrated policy on strategic reserves — of all three 'Fs'.

India once dominated textile trade. It needs institutions to make a mark again



FROM PLATE TO PLOUGH
BY ASHOK GULATI
AND SULAKSHANA RAO

FEW INDUSTRIES have shaped India's civilisation as profoundly as textiles. Long before modern concepts of global value chains emerged, textiles were already woven into India's economic, cultural and commercial fabric. Archaeological excavations at Harappa and Mohenjodaro (dating back 4,500 years) have uncovered spindle whorls, needles and dyeing artefacts, among the earliest evidence of textiles. What began in the Indus valley was sustained through centuries of trade, innovation and craftsmanship.

For nearly two millennia, Indian fabrics clothed consumers across Asia and Europe. During the Mughal period, Bengal became the world's premier centre for cotton textiles. Dhaka produced the legendary muslin prized by royalty across continents, while Murshidabad, famous for its silk and muslin, emerged as one of the 18th century's wealthiest commercial centres. The economic significance of all this was immense. India accounted for one-fourth of global economic output in 1700, according to Angus Maddison. Then the Industrial Revolution happened, powered by James Hargreaves' spinning jenny and other mechanised weaving technologies. These disruptive technologies changed the rules of the game, in this case competitiveness. Competitive advantage stopped being about artisanal skill and became about technology, productivity and economies of scale. India increasingly became a supplier of cotton to Lancashire mills and a market for manufactured textiles.

Independent India's textile journey has been an effort to reverse this. India today possesses a fully integrated textile value chain, from cotton to spinning, weaving, process-

• Global Apparel Market Share of Major Exporters

YEAR	2000	2005	2010	2015	2020	2024
China (%)	17.4%	24.6%	35.3%	37.2%	29.7%	29.4%
Bangladesh (%)	2.2%	2.6%	4.3%	6.1%	8.7%	9.2%
Vietnam (%)	1.0%	1.7%	3.0%	4.9%	6.4%	6.4%
India (%)	3.0%	3.1%	3.1%	3.9%	2.9%	3.0%
World Apparel Exports (USD Billion)	185.0	267.5	342.9	436.5	419.4	521.4

SOURCE: DF FSD

ing, garment manufacturing and exports. The key catch is the employment intensity of the sector. Apparel alone generates 153 jobs for every Rs 1 crore invested, against 27 in automobiles and 14 in steel. Yet its share of global apparel exports has stagnated at around 3 per cent for over two decades even as Bangladesh's rose from 2.2 per cent to 9 per cent, Vietnam's from 1 per cent to 6 per cent. How does a civilisation that once clothed the world end up with 3 per cent of a \$521 billion global apparel market?

This is the question our new book, *Stitching India's Apparel Export Strategy: From Farm to Foreign*, released last week, sets out to answer. The book presents a comparative policy analysis of China, Bangladesh and Vietnam through five pillars of competitiveness: Scale and ecosystem, capital, labour and skilling, clusters and institutions, and market access and trade facilitation.

The lesson from China is one of an ecosystem approach: Vertically integrated clusters, concessional finance and the Sino-sure model of export credit insurance and institutional governance that transformed China into the largest apparel exporter. Bangladesh leveraged Least Developed Country (LDC) preferences through readymade garment (RMG)-specific finance, back-to-back letters of credit, and export processing zones offering plug-and-play infrastructure and one-stop regulatory services. Vietnam aligned foreign investment, industrial parks and free trade agreements (FTAs) into one integrated strategy and industry-academia

India today possesses a fully integrated textile value chain, from cotton to spinning, weaving, processing, garment manufacturing and exports.

The key catch is the employment intensity of the sector

partnerships to attract global manufacturers. Each country followed a different path, but all created integrated ecosystems that enabled firms to compete at scale. Besides all these aspects, one that is very important but much less talked about is the issue of undervalued Chinese currency. Estimates by the IMF, the US Treasury (2026) and Goldman Sachs suggest the yuan has been undervalued by 16 to 30 per cent against the US dollar, acting as an implicit export subsidy.

To realise the government's ambition of \$100 billion in Textile and Apparel (T&A) exports by 2030, India must strengthen these fundamentals of competitiveness across the entire value chain. The first is scale and ecosystem development. The "missing middle" of mid-sized, export-ready firms capable of handling large, fast turnaround orders remains a constraint. Equally relevant is fibre mix. Global apparel trade is dominated by man-made fibre (MMF). India's textile industry remains skewed towards cotton, with a 60:40 cotton-to-non-cotton fibre consumption ratio, in contrast to the global average of 25:75. This mismatch locks India out of the fastest-growing segments such as active-wear, technical textiles and athleisure. The recent rationalisation of the inverted duty structure in the MMF segment must now be complemented by investment in MMF spinning and weaving. Second is capital. India's real interest rates of 6.2 to 8.2 per cent remain significantly higher than those of China (3.1 per cent), Vietnam (around 1 per cent) and Bangladesh (negative real rates). Delays in Goods and Services

Tax (GST) refunds and export incentive disbursements also constrain liquidity. Next is labour and skilling. India's apparel sector is highly dependent on migrant labour, moving from Bihar, Odisha and Jharkhand into production clusters such as Bengaluru, Tiruppur, Surat etc. There is high attrition and absenteeism during the festival season, when assembly lines fall short. In our opinion, a more sustainable model is to move factories to where the cheap labour is. Pearl Global's Muzaffarpur unit in Bihar is a step in that direction. The gains are multiple: Higher female labour participation, lower migration costs and a pathway for labour-surplus states like Bihar to accelerate manufacturing-led growth.

Finally, with the UK-CETA (Comprehensive Economic And Trade Agreement) and EU-FTA, India has achieved tariff parity with competitors such as Bangladesh and Vietnam. But competitiveness extends far beyond tariffs. India needs seamless end-to-end integration of trade processes so exporters can move goods, documents and approvals with minimal compliance costs, speed and predictability.

The reform agenda for India is clear: Build an ecosystem approach to competitiveness. PM-MITRA (PM Mega Integrated Textile Region and Apparel) parks, announced in 2021, embody this approach by integrating the five pillars of competitiveness — scale, infrastructure, finance, skills and market access — within a single institutional framework. Their timely implementation is critical if India is to seize the current window of opportunity in global apparel trade. Complemented by the export promotion mission and the integrated programme for the textiles, these reforms can reposition India as a scale-driven, digitally-enabled, sustainable and women-inclusive global sourcing hub. History tells us that India once clothed the world through craftsmanship. The future will depend on whether it can stitch together an ecosystem where firms, finance, institutions and supply chains move with the speed, scale and predictability that global value chains demand.

Gulati and Rao are distinguished professor and senior fellow, respectively, at ICRIER. Views are personal

Some hard questions to RBI over its growth optimism



ISHAN BAKSHI

IN ITS last meeting, the RBI's Monetary Policy Committee voted unanimously to keep interest rates unchanged and maintain a neutral stance. While the accompanying statements did lay out the rationale for the committee's decision, several issues, beginning with inflation, need to be clarified.

Directionally, inflation is expected to trend upwards through the year. The central bank's own forecasts have pegged inflation at 5.65 per cent in the second half of the year, up from 4.65 per cent in the first half. There is also the possibility of prices surprising on the upside, considering the situation in West Asia and uncertainty over the monsoon season.

The MPC, however, voted to keep the repo rate at 5.25 per cent. This implies that, on a forward-looking basis, real interest rates are mildly negative. As negative rates are meant to stimulate economic activity, this would suggest that policy is accommodative. This sits uneasily with the neutral stance the committee currently maintains. A neutral stance, as the RBI governor had earlier noted, calls for neither stimulating economic activity nor controlling inflation. So, is the MPC implicitly pursuing an accommodative policy while formally maintaining a neutral stance?

Currently, headline inflation is being driven more by food and fuel prices. While there is considerable uncertainty over energy prices, the rainfall this season and the impact of El Niño going ahead, the RBI may well believe that price pressures will moderate towards the end of the year. It may thus view negative real interest rates as transitory, justifying its approach. But as forecasts beyond the second half of this year haven't been released, it is difficult to know for sure. Price pressures may not dissipate. High WPI also does suggest inflation in the pipeline — inflation in manufactured products was 7.18 per cent in the first quarter. The extent of the pass-through of higher input prices depends on the pricing power of firms and domestic demand. Will growing market concentration tilt the balance? And what about the knock-on effects on inflation expectations and wages?

Then there is the issue of how the central bank views the underlying economic growth momentum.

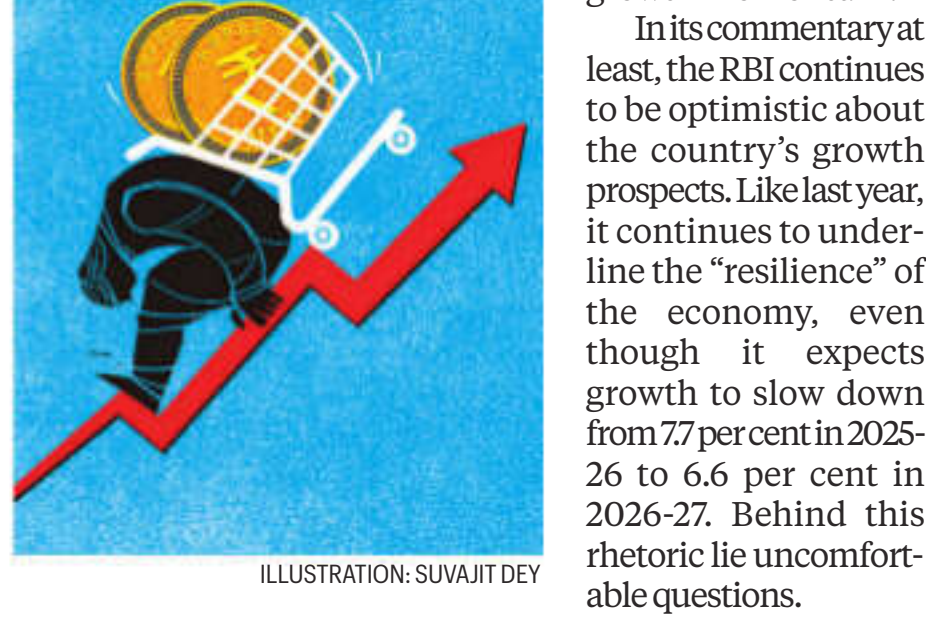


ILLUSTRATION: SUVAJIT DEY

In its commentary at least, the RBI continues to be optimistic about the country's growth prospects. Like last year, it continues to underline the "resilience" of the economy, even though it expects growth to slow down from 7.7 per cent in 2025-26 to 6.6 per cent in 2026-27. Behind this rhetoric lie uncomfortable questions.

The RBI believes that private consumption has displayed "resilience". However, this is at odds with subdued core inflation, which points towards weak demand. "Sustained low level of core inflation indicates that there are no immediate broad-based demand pressures in the economy," an MPC member had noted in the most recent meeting.

The central bank has also argued that fixed investment has maintained its momentum. Yet, its most recent financial stability report has also noted that "private corporate investment remains subdued as reflected in the moderation in the investment to GDP ratio". So, is it the RBI's view that investment activity is still driven by public capex? Do both private consumption and investment need continued support? With inflation edging upwards and growth slowing down, what is the central bank more worried about — growth or inflation?

The calculus, however, does not end there. There is the issue of the rupee and the US Fed.

There have been times when the RBI's interest rate decisions appear to have been guided not by inflation considerations, but more by the desire to defend the rupee. The pressure on the currency, which had eased, now appears to be reigniting following the renewed conflict in West Asia. Will the RBI raise rates if it believes that the measures announced so far are unlikely to ease the pressure on the currency? And what happens if the US Fed also raises interest rates? While inflation in the US has come in below expectations in June, there is a possibility of the Fed raising rates over the next few meetings. If that is indeed the case, will US monetary policy, as it has seemed in the past, influence the RBI's interest rate decisions?

The central bank should clearly spell out its views on these issues.

The writer is associate editor, The Indian Express
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SAROJINI NADIMPALLY

ON JULY 8, the World Health Organisation (WHO) released its Global Status Report on Cancer. The figures are staggering: An estimated 20.6 million new cancer cases and nearly 10 million deaths worldwide each year, that is, more than 26,000 lives lost each day. At current rates, the WHO predicts 35 million annual cases by 2050.

In addition to the scale of the disease, the report highlights an inherent inequity. A woman with breast cancer in a high-income country has an 87 per cent chance of surviving five years, compared to about 42 per cent in a low-income country, a gap that is less due to biology and more due to where she happens to live. While 68 to 94 per cent of wealthier countries have access to the top 20 priority cancer medicines, only 9 to 54 per cent of low- and lower-middle-income countries do.

India's own numbers fit into this picture, albeit not comfortably. With an estimated 15.33 lakh new cases expected in 2024, a Rajya Sabha standing committee noted that roughly one in nine Indians faces a lifetime risk of cancer. The committee has also sought expert recommendations to improve the affordability of cancer care and expand access to diagnostics. According to GLOBOCAN 2022 data, breast cancer accounted for over 1.92 lakh new cases and over 98,000 deaths in India in 2022. More than half of these cases are detected at an advanced stage,

I'm writing this while undergoing cancer treatment myself. What has struck me is not the disease itself but the ordeal of surviving it: Waiting for an appointment, the anxiety, and the arithmetic

making treatment more expensive and less effective. I'm writing this while undergoing cancer treatment myself. What has struck me is not the disease itself but the ordeal of surviving it: Waiting for an appointment, the anxiety, and the arithmetic every patient does, working out which drug she can afford, which treatment package is feasible, and how much her insurance might cover.

A breast cancer patient who cannot afford targeted therapy, immunotherapy, or hormonal therapy is not just facing a market or health-system failure. She is witnessing the slow erosion of a right that the Constitution is meant to guarantee. The same applies to access to essential medicines.

The right to health has long been recognised as integral to the right to life under Article 21 of the Constitution. The Supreme Court has held that a healthy body is the foundation of all human activity; that the right to life includes access to medical care and a decent standard of living; that the state has a duty to make essential medicines affordable.

Litigation over access to medicines, compulsory licensing, or drug pricing should not languish in courts for years, as it often does. In 2022, a retired bank employee with HER2-negative metastatic breast cancer approached the Kerala High Court seeking compulsory licensing of the patented drug Ribociclib under Sections 92 and 100 of the Patents Act, 1970, be-

cause its prohibitive cost made treatment unaffordable. The court directed the Centre to consider compulsory licensing, but the petitioner died before the matter was decided. The High Court converted the proceedings into a *suo motu* case, *In Re Exorbitant Pricing of Life Saving Patented Medicines*, which remains pending despite having been listed for hearing more than 57 times. If delayed treatment is already a rights violation, prohibitively expensive treatment cannot be any less so.

In the 20 years since India aligned its patent regime with TRIPS, it has issued only one compulsory licence: For Bayer's Nexavar in 2012, a drug for kidney and liver cancer, which reduced its monthly supply cost from Rs 2.8 lakh to under Rs 9,000. The 2001 Doha Declaration on the TRIPS Agreement and Public Health, which India helped shape, holds that intellectual property protection should never come at the cost of public health. The tools already exist. What is missing is the willingness to use them.

The WHO has declared this a global emergency. For India, it is a pressing policy issue and a constitutional question. The Rajya Sabha's inquiry into affordable cancer care is a welcome opening. What remains uncertain is the political will to act before it is too late, even for a patient like me who can afford treatment.

The writer is a public-health specialist working on reproductive health and biotechnologies

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Ken-Betwa project: Police detain protest leader fasting for 14 days

Rejecting allegations that Gram Sabha proceedings had been fabricated, Jaiswal said “there were no irregularities in the Gram Sabha records”. On claims that compensation had been sanctioned to a Muslim family despite “no such family residing in Kharihani village, he said the family owned fertile agricultural land there and had been compensated for the acquired land, not for residential status in the village”.

‘PENETRATION INTO TIER-2, TIER-3, RURAL MARKETS IS CRUCIAL FROM AN INCREMENTAL GROWTH VIEW’

Consumer firms eye rural growth as q-comm reshapes urban mkts

Akash Mandal
Mumbai, July 19

AS CONSUMER companies look beyond crowded urban markets for growth, “micro-markets” have emerged as a key strategy. Tata Group-backed Trent, which operates the Zudio brand, highlighted this approach in its annual report for 2025-26, pointing to its focus on expanding across micro markets rather than focusing only on comparative store performance.

These micro-markets, which vary across companies, refer to highly segmented, untapped regions that make it easier for companies to focus their efforts. Tier-1 and tier-2 cities, which have a population with increasing wage growth and aspirational demand, form a large part of the micro-markets for most consumer companies.

Like Trent, many other consumer brands have started shifting their focus towards the hinterlands for growth, as rural demand continues to outpace urban demand, while digitalisation has led to the rapid growth of new-age platforms in bigger cities, saturating the

space for traditional players.

Rural demand growth has continued to outpace urban demand, led by higher agricultural wages, favourable monsoons over the past couple of years that boosted crops, and growth in the rural non-farm economy. This has shifted consumption patterns in these regions, with rural consumers becoming increasingly aspirational. While the El Niño forecast later in the year could affect rainfall and temporarily hamper the momentum, analysts expect rural demand to remain strong going forward.

Many consumer brands have acknowledged this shift and made efforts to expand into tier-2 and tier-3 cities, a trend also reflected in their earnings.

In their earnings announcement on July 11, Avenue Supermarts, which operates the retail chain D-Mart, noted that its stores in non-metro cities performed much better than the older stores in bigger cities in the April-June quarter.

“In large metros, growth in older stores, which have significantly higher revenue per square foot, was flat this quarter. While stores in non-metros continue to grow well,”

● CONSUMPTION PATTERNS BEYOND METROS

RURAL DEMAND growth has continued to outpace urban demand, led by higher agricultural wages, favourable monsoons, growth in rural non-farm economy

WHILE THE El Niño forecast could affect rainfall and temporarily hamper the momentum, analysts expect rural demand to remain strong

MICRO-MARKETS, which are in focus, are highly segmented, untapped regions that make it



easier for firms to focus their efforts. Tier-1 and tier-2 cities form a large part of micro-markets

said Managing Director and Chief Executive Officer Anshul Asawa. These stores played a major part in the company’s 15% revenue growth during the quarter, the management said.

Other retail conglomerates such as Tata Group’s Titan and Reliance Group’s retail ventures have also expanded to the hinterlands, with a large chunk of the new stores added every quarter placed in regional hubs and smaller locations.

The same is true for quick-

service restaurant companies like Jubilant Foodworks and Devyani International, which run the Indian outlets for international brands like Domino’s, KFC, and Pizza Hut. These companies had earlier focused on increasing their store density across urban hubs so that a Domino’s or a Pizza Hut is present in every locality in a big city.

Traditional FMCG players, which earlier focused on specific rural hubs with dense populations, have expanded further

into the hinterlands on the back of digitalisation and improving trade channels. After the Covid-19 pandemic, companies such as Hindustan Unilever (HUL) and Nestle India have launched and optimised specialised programmes to aid this.

“If we look at any FMCG company pre-Covid, a large part of the focus was on the urban markets. A lot of the big names such as HUL and Nestle already have almost 100% coverage in the urban market. Thus, penetration into tier-2, tier-3, and rural markets becomes important from an incremental growth perspective,” said Ronak Shah, an analyst tracking the consumer sector at Equirus Securities.

The shift in focus to smaller and untapped markets has been gradual and was driven by several factors. A major reason behind this shift is firms chasing incremental growth. Another key reason is the high competition provided by new-age quick commerce platforms such as Zepto, Swiggy’s Instamart, and Zomato’s Blinkit, with most of their sales coming from metropolitan cities.

FULL REPORT ON
WWW.INDIANEXPRESS.COM

Healthcare crisis in US, regulatory curbs weigh on Indian IT companies

Akash Mandal
Mumbai, July 19

EARNINGS FOR most Indian IT companies so far have shown resilience on several fronts, even as the demand outlook remains clouded by factors such as geopolitical uncertainty, client budget cuts and rising AI spends.

While each IT major has its own set of clients spread across various verticals, all these players have singled out one segment that has underperformed across the board — the healthcare and lifesciences vertical in the Americas.

For example, revenue from Wipro’s Health vertical, which accounts for 14% of its overall revenue, fell 2.6% sequentially in Q1. In its post-earnings conference call on Thursday, Chief Executive Officer (CEO) Srinu Palia said budgets of US-based healthcare providers and insurers have remained flat due to pressures in the US healthcare ecosystem, with these companies also looking to increasingly spend towards AI.

“Their budgets have been flatish for us, and in some places we have seen negative growth. Also, most of the budgets right now are being reallocated to some kind of discretionary spending, but towards AI,” Palia said.

Other IT majors have also seen revenue from the segment decline sequentially in Q1 — by 1% for TCS and 0.5% for HCL Technologies. Tech Mahindra was the only exception, with the segment registering positive growth of 2.6%.

Why healthcare underperformed

A large chunk of the healthcare and lifesciences verticals

● Healthcare revenue drags IT majors in Q1

Company	Revenue change from health	Rev share (QoQ change)
TCS	-1.0%	10.3% (-10 bps)
Wipro	-2.6%	14.0% (-20 bps)
HCL Tech	-0.5%	14.0% (-20 bps)
Tech Mahindra	2.60%	7.3% (unchanged)

Note: Health refers to Healthcare and Life Sciences. All changes are QoQ, in constant currency terms
Source: Exchange filings

of Indian IT majors comprises US-based healthcare providers, insurers, life sciences companies and medical device manufacturers. The US healthcare system has faced an affordability crisis since the Covid-19 pandemic, prompting the government to tighten scrutiny and introduce measures that have pressured companies in the sector over the past few years.

Some of the measures include capping prices of critical drugs and aggressively squeezing medicare-related costs, which is positive for citizens but have sharply hit the margins of companies in the sector.

“The government crackdown has reduced the discretionary spending power of US health companies and payers. What has made it worse for Indian IT companies is that these US clients have now focused their limited spending ability towards AI and automation to reduce costs in that way as margins remain squeezed,” an analyst tracking the IT sector at a domestic brokerage said.

Traditionally, healthcare companies in the US approached Indian IT firms for jobs such as server maintenance, back office operations, and compliance. However, they are now spending heavily on AI, in which the Indian ma-

jors face much more competition from global AI companies.

‘Spending shifts towards AI’

“It is very difficult to say from here what the issue is. It might be a sector-specific issue, or maybe a client-specific one. But discretionary spending, where you experiment on new stuff, is definitely declining in favour of AI spending,” said Sunny Agrawal, head of research at SBI-CAPS Securities.

These factors have led to the healthcare and life sciences vertical being on a declining trend over the past year across many Indian IT majors. For instance, Wipro has seen their revenue from the segment falling sequentially in three of the last four quarters. HCL Tech saw its revenue from the segment fall 1.8% in FY26.

However, some of these companies expect performance from the segment to improve in the upcoming quarters. TCS Managing Director and CEO K Kirishivasan said he expects the vertical to pick up in the Q2 earnings. Wipro’s Palia said while companies in the American healthcare sector focus on AI, he expects new opportunities such as AI implementation and compliance to also arrive.

FY27 growth seen 6.5-6.8%: Deloitte

Press Trust of India
New Delhi, July 19

DELOITTE INDIA on Sunday projected India’s economy to grow at 6.5-6.8% in the current fiscal, with growth expected to strengthen in the second half of the year supported by festive demand, monetary easing, and a gradual stabilisation in global conditions.

In its latest edition of Economic Outlook report, Deloitte said India entered 2026 in a Goldilocks phase, with macroeconomic fundamentals appearing unusually well balanced, but geopolitical developments altered the global landscape with tensions in the West Asia disrupting critical shipping routes, triggering volatility in commodity prices and weakening investor sentiment.

This resulted in a wider trade deficit, sustained capital outflows, and a sharp depreciation of the rupee against the US dollar within a matter of weeks.

Against this backdrop, the RBI had last month lowered India’s GDP growth estimates for current fiscal to 6.6%, from 6.9% estimated earlier. GDP grew 7.7% in the previous (2025-26) fiscal.

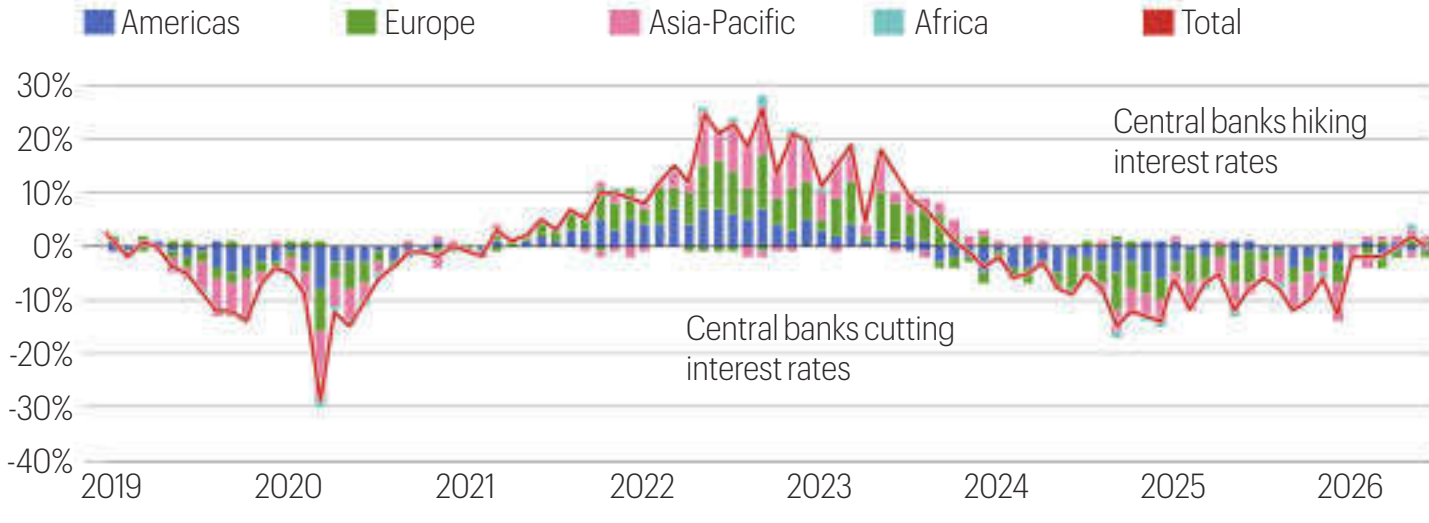
Deloitte India Economist, Rumki Majumdar said the global environment has become considerably more uncertain.

“While recent geopolitical developments and the RBI’s policy measures may help cushion some of these risks, weather-related uncertainties, particularly the impact of El Niño on agricultural output and food prices, remain an important downside risk,” she said.

“Deloitte expects India’s economy to grow between 6.5% and 6.8% in FY27. Growth is likely to remain moderate in the first half before strengthening in the second half, supported by festive demand, monetary easing, and a gradual stabilisation in global conditions,” Majumdar said.

● ‘RATE HIKES WILL NOT REOPEN THE STRAIT OF HORMUZ’

Central banks are having to be more careful than usual given the growth, inflation uncertainty caused by the geopolitical and trade shocks, according to a report by Moody’s Analytics.



SOURCES: BANK FOR INTL. SETTLEMENTS, MOODY’S ANALYTICS

● Higher inflation, even if temporary in nature, strengthens the case for tighter monetary policy, especially at a time when the AI boom has kept growth strong and asset prices elevated.

● But raising interest rates to curb inflation is less effective

now than say, in 2022, as demand is already weak across much of the world, contended Moody’s Analytics.

● “...most central banks have not eased much since the 2022 tightening cycle, and rates in many countries sat in contrac-

tionary territory before the West Asia flared up. Much of the easing was meant to happen this year. Instead, the conflict and the murky outlook have forced many policymakers to hold steady or nudge rates higher.”

SOURCE: MOODY’S ANALYTICS

India, Spain advance talks on UPI-Bizum interoperability

New Delhi: India and Spain have agreed to fast-track discussions on enabling interoperability between India’s Unified Payments Interface (UPI) and Spain’s Bizum digital payments platform, the Commerce Ministry said on Sunday.

The two countries have also discussed ways to strengthen investment facilitation, improving mobility of professionals and promoting joint investments in Latin America, leveraging Spain’s strategic position as India’s gateway to Europe and Latin America.

These issues came up for discussion during the recent visit of Commerce and Industry Minister Piyush Goyal to Spain.

“Both sides also agreed to advance technical discussions for interoperability between India’s UPI and Spain’s Bizum digital payments platform,” he said. The two sides have also identified key opportunities in automobiles, pharmaceuticals, chemicals, textiles, renewable energy, tourism and infrastructure. PTI

‘Semicon 2.0 to drive investment for chips’

New Delhi: Provisions under Semicon 2.0 to provide incentives to Indian chip-making firms against equity are expected to drive largescale investment required for the development of advanced chips for technology such as artificial intelligence, a senior government official has said.

Under Semicon 2.0, which has a budget of Rs 1.27 lakh crore, the central government

has prioritised chip design by Indian companies.

In an interview to PTI on Saturday, Ministry of Electronics and IT Secretary S Krishnan said advance chip design may need thousands of crores of investments.

“Under design-linked incentive scheme, you get only about Rs 15 crore. Now, to design high-end chips you need Rs 1,000 crore or more,” he said. PTI

● INDIA-AUSTRALIA URANIUM DEAL: THE COUNTRY IS SECURING URANIUM AS IT AIMS TO EXPAND NUCLEAR POWER CAPACITY TO 100 GWE BY 2047

Why domestic reserves are not enough to fuel India’s nuclear ambitions

Pratyush Deep
New Delhi, July 19

EARLIER THIS month, India secured a long-term uranium supply deal with Australia, marking the first major outcome of the India-Australia Civil Nuclear Cooperation Agreement signed nearly 12 years ago.

While the volume of uranium oxide, or yellowcake, to be supplied and the duration of the agreement have not been disclosed, the deal follows similar long-term uranium supply agreements signed with Canada and Kazakhstan earlier this year, underscoring New Delhi’s efforts to diversify and secure fuel supplies for its planned expansion of nuclear power generation.

The agreement comes as India pursues an ambitious tar-

get of expanding its civil nuclear power capacity to 100 gigawatt-electric (GWe) by 2047 and, last year, opened the tightly regulated civil nuclear sector to greater private participation.

Uranium, a common element found worldwide, must undergo several processing stages before it can be used as nuclear fuel. These include mining and milling, conversion, enrichment (where required), and fuel fabrication. But where does India currently source the uranium that fuel its nuclear reactors? And what role do the domestic uranium reserves play in meeting the country’s growing demand?

India needs more uranium supply

For the country’s indigenous pressurised heavy water reactors (PHWRs), which run

on natural uranium, fuel requirements are met through a combination of production from state-owned Uranium Corporation of India Ltd (UCIL) mines and imports. Enriched uranium used in the two light water reactor (LWR) variants — Tarapur Nuclear Power Plant and Kudankulam Nuclear Power Plant — is entirely imported.

The domestic uranium oxide reserves currently stand at around 4.3 lakh tonnes, spread across 47 uranium deposits located in Andhra Pradesh, Telangana, Jharkhand, Meghalaya, Rajasthan, Karnataka, Chhattisgarh, Uttar Pradesh, Uttarakhand, Himachal Pradesh and Maharashtra. UCIL has the mandate to mine and process uranium ore in the country and operates major mines in Jaduguda and Tur-



Uranium must undergo several processing stages before it can be used as nuclear fuel. REUTERS

amdih in Jharkhand and Tummalapalle in Andhra Pradesh.

However, the uranium requirement for only 2.4 GWe out of the country’s total 8.7 GWe capacity is currently met

● URANIUM DEFICIT

DOMESTIC URANIUM oxide reserves stand at around 4.3 lakh tonnes, spread across 47 uranium deposits located in Andhra Pradesh, Telangana, Jharkhand, Uttar Pradesh, Uttarakhand, Himachal Pradesh among others

BUT URANIUM requirement for only 2.4 gigawatt-electric out of the country’s total 8.7 gigawatt-electric capacity is currently met from domestic sources while the rest is met through imports

from domestic sources while the rest is met through imports, a committee constituted by the Ministry of Power to prepare a roadmap for 100 GW nuclear capacity by 2047

noted in June 2025.

“Thus, to meet the target of 100 GW of nuclear capacity in 2047, India would need to augment uranium supplies and also have commensurate processing and fabrication capabilities,” it added.

The committee estimated that in 2047, the country will need an additional 8,029 tonnes of natural uranium and 1,045 tonnes of enriched uranium annually.

NSG waiver, global uranium partnerships

The country’s ability to import uranium for its civil nuclear reactors was unlocked in 2008, when the Nuclear Suppliers Group (NSG) granted it a ‘clean’ waiver, making it the only non-signatory to the Nuclear Non-Proliferation Treaty (NPT) to

enjoy such trading privileges.

Government data shows that between 2008-09 and 2024-25, India imported a total of 18,842.60 metric tonnes (mt) of uranium in the form of uranium ore concentrate, natural uranium dioxide pellets and enriched uranium dioxide pellets for reactors operating under International Atomic Energy Agency (IAEA) safeguards.

In 2024-25, nuclear power plants generated 56,631.48 million units (mu) of electricity, of which 39,180 mu was produced using imported uranium, according to government data. The estimated requirement of imported natural uranium for operating and proposed safeguarded PHWRs is around 9,000 metric tonnes of uranium during 2025-2033.

India currently imports uranium from Russia and Uzbekistan. Shipments from Canadian mining company Cameco Corp., under the \$2.6-billion supply agreement signed in March, are expected to begin next year. Kazakhstan is also expected to emerge as one of India’s most significant uranium suppliers after Kazatomprom, the world’s largest uranium producer, agreed in February to sell a substantial volume of uranium to India. Both Cameco and Kazatomprom were among India’s major uranium suppliers until 2020-21.

Apart from these countries, France supplied uranium ores and concentrates to India in 2025-26, while similar imports were received from the US in 2024-25, according to trade data.

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• ECONOMICS

Poor monsoon’s impact on India’s agri import bill



HARISH DAMODARAN

IN 2025-26 (April-March), India imported a record 16.9 million tonnes (mt) of vegetable oils and 1.1 mt of raw cotton, valued at \$19.5 billion and \$1.9 billion respectively. Further, it imported nearly 6 mt of pulses — the highest after 2024-25 and 2016-17 — worth \$3.6 billion (*table 1*).

The current fiscal could see even these records broken, thanks to El Niño. The impact of this climate phenomenon — an abnormal warming of the central and eastern Pacific Ocean waters off the coasts of Peru and Ecuador, associated with dry weather across India, Southeast Asia and Australia — is already being felt.

Rainfall for the country in June was 38% below the normal long-period average for the month. While July so far has registered a lower 9.9% shortfall, the cumulative deficiency for the four-month southwest monsoon season (June-September) as on July 19 is still at 23.8%.

Kharif sowing lag

The lack of rain has affected plantings of most *kharif* (monsoon-cultivated) crops. While overall sown area till July 10 was 16% below that for the same period of 2025, the decline was even more in pulses (23.3%) and oilseeds (21%).

Within pulses, the year-on-year acreage drop was 30.3% for *arhar* (pigeon pea) and 29.7% for *urad* (black gram). In oilseeds, it has ranged from 16% for soyabean to 34% for groundnut and 46% for sesamum. The progressive area sown under cotton, too, has fallen 15.3% from last year’s corresponding coverage (*table 2*).

“Rainfall has been weak and also scattered. There are areas within 20-25 km of the same *taluka* (sub-district) that have received showers and those that haven’t,” said Nitin Kalantri, a leading dal miller from Latur in Maharashtra.

The monsoon’s revival seen during the first week of July should help somewhat close the above gaps and may be reflected in the Agriculture Ministry’s *kharif* sowing

• How India’s imports of these three have soared

Year (Apr-Mar)	Vegetable Oils		Pulses		Raw Cotton*	
	Quantity	Value	Quantity	Value	Quantity	Value
2016-17	140.07	10892.75	66.09	4244.13	5	946.88
2017-18	153.61	11637.48	56.08	2908.33	4.69	979.32
2018-19	150.19	9890.32	25.28	1140.76	2.99	633.05
2019-20	147.22	9672.88	28.98	1440.09	7.44	1328.42
2020-21	135.4	11089.12	24.66	1611.72	2.31	385.89
2021-22	142.78	18991.62	27	2228.95	2.24	559.55
2022-23	157.45	20837.7	24.96	1943.89	4.52	1438.69
2023-24	155.2	14871.66	47.39	3746.78	2.52	598.66
2024-25	164.13	17366.26	72.56	5477.28	6.01	1219.32
2025-26	169.4	19488.41	59.64	3572.89	10.73	1888.79

QUANTITY IS IN LAKH TONNES AND VALUE IS IN \$ MILLION; *INCLUDES WASTE.
SOURCE: DEPARTMENT OF COMMERCE

data update on Monday.

“Sowing of pulses, including *arhar* (a 5-6 months crop), can take place till July-end. What we need is good rain over the next two weeks,” added Kalantri.

The same goes for cotton, where the sowing window in the main growing belts of central and southern India usually shuts by mid-July. This can be extended till the month-end, if the monsoon is delayed or erratic.

“Cotton doesn’t need too much rain. Also, the price sentiment for cotton is good this time, with many farmers keen to shift from rice and maize,” M Prabhakar Rao, chairman and managing director of the Medchal (Telangana)-based Nuziveedu Seeds Ltd, pointed out.

According to him, less rainfall would be a problem more from the standpoint of pink bollworm (PBW) infestation, which can bring down crop yields: “PBW is a monophagous pest that feeds exclusively on cotton. Frequent rains help in disrupting the mating cycle of the adult moths, besides drowning the pupae in the soil. Prolonged dry weather is conducive for uninterrupted mating, egg-laying and the growth of larvae that bore into the developing bolls.”

El Niño worries

While the monsoon rainfall deficit is

• Kharif acreage growth

Pulses	-23.32	Oilseeds	-21.01
Arhar /Tur	-30.29	Soyabean	-15.98
Moong	-10.63	Groundnut	-33.99
Urad	-29.72	Sesamum	-46.03
Moth Bean	-28.1	Sunflower	62.26
Cotton	-15.34	Coarse Grains	-22.47
Rice	-8.64	Maize	-19.54
Sugarcane	1.52	Bajra	-26.58
Jute & Mesta	1.95	Jowar	-23.64
All Kharif Crops	-16.03	Ragi	-16.22
		Small Millets	-29.27

(%, YEAR-ON-YEAR), AS ON JULY 10
SOURCE: DEPT. OF AGRICULTURE & FARMERS’ WELFARE

quite significant now, the worry is more from a strengthening El Niño in the months ahead.

The US National Oceanic and Atmospheric Administration (NOAA) has forecast an 81% probability of the present El Niño intensifying into a “very strong” event during October-December, with a 97% chance of it persisting through the spring months till March-April.

Given that El Niño is known to not only suppress rainfall, but also raise temperatures, it could translate into a relatively short and warm upcoming winter. That can potentially hurt the ensuing *rabi* (winter-spring) season crops such as wheat, rapeseed-mustard, *chana* (chickpea), *masoor* (red lentil) and potato as well.

The brunt of all this would be borne by oilseeds, pulses, millets and cotton, which are predominantly rainfed crops, unlike rice, wheat or sugarcane. The current developing El Niño — which, NOAA believes, may turn out to be “among the largest...events in the historical record going back to 1950” — can result in India’s vegetable oil, pulses and cotton imports also scaling a new peak.

The comfort factors

On the positive side, stocks of rice and wheat in government godowns, at 68.3 mt and 53.4 mt respectively on June 1, were way above the corresponding minimum

Worries ahead

While the monsoon rainfall deficit is quite significant now, the worry is more from a strengthening El Niño in the months ahead.

Given that El Niño is known to raise temperatures, the upcoming winter could be relatively short and warm. That can potentially hurt the *rabi* crops too.

• SPACE

Race for Low Earth orbit: What Vikram-1 launch means for India

Arav Shah

New Delhi, July 19

INDIA’S FIRST privately developed launch vehicle, Vikram-1, placed several technology demonstration satellites in Low Earth orbit (LEO) on Saturday morning.

The launch from Sriharikota by Hyderabad-based Skyroot Aerospace — India’s first space unicorn — makes India only the third country globally to have private launch capability after the US and China. This comes after the government opened up the space sector to private companies in 2020 and formalised the framework under the Indian Space Policy of 2023.

Low Earth orbit

LEO is the zone immediately surrounding Earth and is typically defined as the region between 160 km and 2,000 km above our planet’s surface. Satellites in LEO generally take 90-120 minutes to circle Earth. This proximity makes this orbit preferable for satellites meant for communications as well as Earth observation and imaging because it helps reduce signal latency.

“Today, orbital slots in the LEO work on a first-come-first-served declaration. The only obligation a new entrant has is to not interfere with the previous occupants. This



Vikram-1 lifts off from Sriharikota, Andhra Pradesh, on Saturday. PTI

essentially makes it a literal gold rush,” Ashwin Prasad Rao, staff research analyst at the Takshashila Institution’s Outer Space programme, told *The Indian Express*.

Under the 1967 Outer Space Treaty, no nation or corporation can legally claim ownership over an orbit.

“What states are actually competing for is access to preferred orbital shells and advantageous radio-frequency slots managed by the ITU,” Khyati Singh, research analyst at the Manohar Parrikar Institute for Defence Studies and Analyses, told *The In-*

Partially reusable

Although India doesn’t have a reusable launch vehicle yet, ISRO’s partially reusable Next Generation Launch Vehicle is projected to be operational in the mid-2030s.

dian Express. The International Telecommunication Union (ITU) is the United Nations agency managing radio frequencies and preventing harmful interference between two or more satellites.

An “orbital shell” is a layer of satellites at the same altitude (how high the satellite orbits above Earth’s surface) and orbital inclination (angle of the satellite’s orbit relative to the equator). Once a specific orbital shell is occupied, subsequent parties are expected to coordinate around the incumbents or migrate to clearer pastures.

So, this framework incentivises speculative “squatting”, leading to countries designing and deploying satellite “megaconstellations” to encircle as much space as possible — like China’s Guowang and Qianfan satellites.

A promising start

Vikram-1’s success represents a massive leap for Indian private aerospace capability. It ensures New Delhi has assured access for launching small satellites, which are currently the most “underserved segment of the market,” Singh said.

But its true strategic value lies in creating sovereign “surge capacity.” “This grants India the ability to rapidly rebuild or increase the number of intelligence, surveil-

lance, and reconnaissance satellites on demand, without relying on ISRO, which is vital in an increasingly contested Indo-Pacific region,” she said.

Securing this orbital access directly feeds into national security. “It is a huge information warfare tool,” Rao said, pointing to the Starlink satellite system’s role during the Iran war. He, however, cautioned against the “current model of overreliance on a few external providers”, citing the increasing risk of security concerns.

Still some way to go

India is still far behind in the LEO race. To counter this vulnerability, India needs to ramp up its launch frequency. But this faces major structural bottlenecks.

“Two of the biggest roadblocks right now are the lack of scale that the sector is going through. The other is ISRO’s elongated timelines for India’s own reusable rocket which is one of the most critical components in ensuring frequent launches into the LEO,” Rao said.

Building this ecosystem requires immense capital. While the government recently introduced a Rs1,000-crore venture capital fund to support early-stage space startups, an increase in direct procurement volume is the need of the hour.

• GLOBAL

US, Iran trade lethal strikes again: Their motives, lessons

Shubhajit Roy

New Delhi, July 19

THE US and Iran have attacked each other viciously in the last one week, blowing multiple holes into the already tattered Memorandum of Understanding (MoU) they signed on June 17. On Saturday, two US military personnel were killed in Jordan. The US has retaliated, raising the spectre of an all-out war once again.

The MoU had hinged on the capability Iran had demonstrated to control the Strait of Hormuz. The recent tensions hinge on the US attempting to demonstrate that it can loosen Tehran’s chokehold on the strategic international waterway, through which 20 per cent of the world’s energy supply flows.

Takeaways of war for both

Iran has said that US strikes have killed at least 50 people and wounded more than 500 in the post-MoU round of hostilities. The MoU had promised major economic relief to a country crippled by sanctions. Yet, Iran’s unwillingness to back down is because of two factors — the conviction that the US cannot be trusted, and the imperative to maintain control over the Strait of Hormuz, its biggest leverage.

Thus, Iran has continued to escalate wherever it sees the need, attacking US military bases in its neighbouring countries. On July 18, the new Ayatollah, Mojtaba Khamenei, posted on X that the “repeated breaches of the agreement” show that the “signature of the US President is utterly worthless and devoid of credibility.”



A bridge destroyed after a strike in southern Iran. AP

• INDIA’S CONCERN

A return to war will bring serious challenges at a time the Indian economy is already facing headwinds.

New Delhi is urging both parties to return to dialogue and diplomacy, and to not put the global economy on a slowdown.

high-speed, manoeuvrable missiles at US bases in the region, highlighting the threat that the advancement of Iran’s ballistic missile programme poses to US security interests, according to ISW.

The ISW said Iranian leaders focused on missile accuracy after the 12-day war in June 2025. Iran had struggled with relatively poor missile accuracy in its strikes against Israeli military targets in March and April 2026 too, which likely pushed it to deploy cluster munitions at a greater scale in order to maximise damage.

For the US, the experience has been a re-hammering of the lessons it learnt after the wars in Afghanistan and Iraq — that too much firepower, poorly applied, only produces more enemies. A major goal for the US now is to ensure Hormuz remains free for movement, like it was before it, along with Israel, attacked Iran on February 28.

Stakes for India

For Delhi, a return to open war means more lives at risk, as many Indians live and work in West Asia, apart from the large population of Indians that make up ship crews. Fourteen Indians have died since the war began on February 28, while two more are missing. New Delhi has protested with both Iranians and Americans over the loss of lives — mostly of seafarers and some Indians living and working in the Gulf.

War will also bring serious economic challenges. The Indian economic situation is facing headwinds with the rise in energy prices, which has had a debilitating impact on its markets. The inflationary impact of West Asian developments is already visible.

• ARTIFICIAL INTELLIGENCE

In China’s creation of new global AI body, a push for lead role in governance

Amitabh Sinha

New Delhi, July 19

SINCE 2018, the World AI Conference (WAIC) has been held annually in Shanghai to showcase Chinese advancements in artificial intelligence (AI). What has largely been an industry event has attracted global attention this year, thanks to the participation of Chinese President Xi Jinping. Several heads of state and UN Secretary-General Antonio Guterres were also present upon his invitation.

Xi’s presence led to the event doubling as a platform for presenting an alternative thought process and framework for global AI governance — an evolving subject that will be central to how the technology grows and who benefits from it.

What China is offering

At the core of this alternative perspective is the creation of a new international institution called World AI Cooperation Organisation, or WAICO. It is meant to ensure that

developments in (and benefits of) AI are accessible to everyone, even developing countries that lack the resources or capability to build their own AI systems from scratch. It also commits to “extensive consultation and joint contribution for shared benefit”.

WAICO was proposed at last year’s conference and has now been made operational. Along with China, 29 countries (including Pakistan and Russia) signed an agreement for its creation. It is being projected as a more “inclusive” alternative to the AI governance frameworks proposed by Western countries, which heavily emphasise ethics, values, safety and responsibility.

In this context, Xi’s keynote speech on the opening day of the conference about the need to guard against “overstretching” of national security concerns is being widely quoted.

“AI development should not be a solo performance by a single country, but a symphony of international cooperation... we should jointly oppose overstretching the national security concept in the field of

AI and placing one country’s security over that of others,” he said.

Cornering technology

It is undeniable that AI, like other powerful technologies with wide-ranging impacts, has a very high risk of potential misuse. Nuclear technology, for instance, produces clean electricity, has applications in healthcare, agriculture and research, and is also behind the most devastating weapons known to the world.

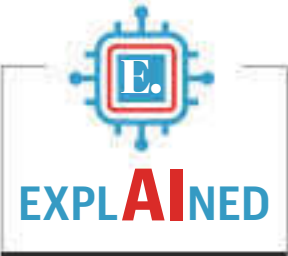
The misuse of AI is seen to have even more profound consequences for humanity. It is no surprise, therefore, that in conversations on governance or regulatory frameworks, issues of safety, ethics and responsibility are often emphasised.

But there is also a growing concern that these legitimate issues might become a cloak for technology denial or exclusion of

countries. Such concerns are not unfounded. Historically, issues concerning environmental standards, child labour, or intellectual property protection have been used to deny certain technologies, or impose trade restrictions and other sanctions.

Case in point, the sudden shutdown of the US-based Anthropic’s new Mythos 5 and Fable 5 AI models last month, following a US government directive to suspend its access to all non-US nationals.

Since a bulk of AI development is happening in the United States, it is also spearheading public debates on governance and regulation, alongside like-minded countries in Europe and elsewhere. Globally, a disparate set of conversations on governance frameworks is ongoing. There is the UN-initiated Global Dialogue, and the recent AI Summit in New Delhi, which saw a group of countries facilitating these discussions.



A weekly column decoding AI’s new frontiers, how they shape the world and our place in it

Idea Exchange

MONDAY, JULY 20, 2026

NEWMAKERS IN THE NEWSROOM

WHY PHILIPP ACKERMANN

Year 2026 marks the 75th anniversary of the establishment of diplomatic relations between India and Germany. Given its key role in the EU and the strength of bilateral relations, Germany is one of India's most significant partners in Europe. German ambassador Philipp Ackermann, who has served in India from 2022 to 2026, and was posted in Delhi from 2007 to 2010, has witnessed the changing arc of the relationship. He has witnessed strengthening of bilateral ties and navigated the challenges — wars in Ukraine, Gaza and Iran and the volatility in geopolitics. As he heads to China, he is best placed to reflect on Indo-German ties in an increasingly uncertain world

Shubhajit Roy: You've seen India-Germany relations over two different political eras in India — the UPA years and the Modi years. As India and Germany mark 75 years of diplomatic ties and 25 years of strategic partnership, how has the relationship evolved?

There has been a clear shift of attention in India towards Europe and Germany, and in Germany and Europe towards India. It felt as if I were in a magnetic field. All of a sudden, there was such renewed interest in what happens in Europe and what Europe can contribute to India's development — and, at the same time, in how India can be a partner for Europe in an unruly world. That was just stunning. For us, the most important thing is perhaps India's Free Trade Agreement (FTA) with the European Union, which will change our approach to doing business with and in India. That it finally came into being after such a long time is a milestone. So what we have seen is, in many ways, a very granular relationship with India — a strategic relationship that all of a sudden opened up new areas, for example, defence and migration. We finally agreed to lift transit visas, something that for a long time nobody had been willing to consider. There is such goodwill on both sides that you can do a lot with it. I look back on four years of a renewed and consolidated partnership, a more multi-layered partnership, and that is a very satisfying feeling. It is also the global environment, which is really not the easiest. But in this volatile, unruly global environment, this is a partnership that will play a big role in the future.

Shubhajit Roy: Could you talk about the FTA negotiations? What were the biggest challenges with India, and how were they unlocked?

When I first came here in 2007-2008, these negotiations had just started. After some time, both sides found it all too complicated, and the talks were put in the deep freeze for 10-12 years. Then, two or three years ago, people sat together again. At a time when free trade is no longer popular, we actually still want it. We like free trade. So it was a signal that we should sit together and try to work out a free trade agreement between India and Europe. When the EU does FTAs, it insists on very high social and environmental standards; then the carbon question came up. And the Indian side said: you have to come up with a tailor-made solution for us. I think the European Commission understood this. When the political pressure is there, the negotiating teams eventually agree. The big problem for Europe in free trade is agriculture. Very wisely, in this agreement, agriculture was largely excluded. Wines and spirits are included, but many products, where Indian agriculture is so different from European agriculture, were left aside. That was a good decision; it will make for an easier debate in the European Parliament because many there see free trade exclusively through the agricultural lens. With India, we also worked out how to word certain social and environmental standards so that they are not harmful to the Indian economy. And for Germany, the priorities were car-making and machinery. We were very satisfied with the outcome. After a long and sometimes tough negotiation period, everybody is quite happy with it.

Shubhajit Roy: A significant element of the Indo-German relationship has been the opening of doors to Indian students, businesses and skilled professionals. But with the rise of the far Right in German politics, do you see this accomplishment being put at risk in the future?

There is no doubt that migration, as in every other European country, is a contentious issue in Germany and a lot of people feel rather uneasy after the two big waves of migration: Syria in 2015-16 and Ukraine in 2022. But at the same time everybody who is exposed to the reality of Germany understands that there are areas where the country urgently needs talent. You need engineers, nurses and caregivers, and you need apprentices who start out in a craftsman's workshop or as a butcher, a roofer or a baker. The federal government has understood that we need fair, legal migration channels for people who want to move to Germany. When you go to any hospital in Germany now, whether in the countryside or in the cities, you will find a lot of non-German personnel. Even when you talk to AfD (Alternative for Germany) politicians in the German Bundestag, they will concede, albeit grudgingly, that they too see the need for skilled labour in many areas.

Germany has a shrinking population. People will come to understand that these fair and legal immigration channels are good for the nation. There is a new Hindu temple in Berlin whose *gopuram* is the tallest in Europe; it took 20 years to build. I was very impressed by how well the Indians there had integrated into the German labour market after their studies. There is one problem, though: the German language. Beyond Berlin, getting by with English is a problem — German is an essential skill. But generally, Indian migration has been a success in Germany.

Shubhajit Roy: The war in Gaza has tested and stretched Germany's support for Israel, which is rooted in post-war guilt. Is that support waning?



ON RUSSIA'S INTERFERENCE

Russia wants to invade other countries in order to enlarge its territory. It is interfering heavily in every European country through cyber warfare. We have to be in a better position to defend ourselves



ON INDIA-EU FREE TRADE AGREEMENT

For us, the most important thing is perhaps India's Free Trade Agreement with the EU, which will change our approach to doing business with and in India. That it happened, is a milestone

'Indian migration is a success in Germany... I am impressed how Indians have integrated into the German labour market'

As he prepares to leave Delhi for Beijing after four years as Germany's Ambassador to India, Philipp Ackermann on 75 years of diplomatic ties, migration, Europe's security after Russia's invasion of Ukraine, and what still surprises him about India. The session was moderated by Shubhajit Roy, Diplomatic Editor, *The Indian Express*



German ambassador Philipp Ackermann (right) in conversation with Shubhajit Roy, Diplomatic Editor, *The Indian Express* GAJENDRA YADAV

How has the war complicated Germany's relationship with Israel?

This goes to the very foundations of modern post-war Germany. Support for Israel among the population is really shrinking. When you talk to young people now, an increasing number of people are critical of the Israeli government. This includes major political parties — the Social Democrats, the Greens — who were very critical of the current government. Chancellor Friedrich Merz stopped delivering arms to Israel when he felt that the Gaza war got out of hand. So there is an incremental change underway. There will never be a moment when Germany is not defending the right of existence of Israel. But in Germany, you now have a movement that is very strongly pro-Palestinian. That also has to do with migration: we have a lot of Syrian, Palestinian and Arab migrants in Germany, and that leads to conflicts and frictions. But when you read the government's policy statements on Israel, the main thing we now hear is harsh criticism of the settlements in the West Bank, which we consider illegal. I sense a certain cautious movement away from absolute support.

Shubhajit Roy: After Russia invaded Ukraine, you were forced to rearm. But Russia has not collapsed and is still fighting Ukraine. It has even got a fresh lease of life with the Strait of Hormuz closed.

Nobody wants the collapse of Russia. We had a long-standing relationship with Russia — there was a NATO-Russia Council, and Germans were explicitly reaching out to Russians, not only for business reasons but also for historical ones. We felt a responsibility towards a country that had suffered enormously during the Second World War. But what we woke up to one day was that Russia is an imperialist country that wants to invade other countries in order to enlarge its territory. That we cannot accept in Europe because we are built on the clear understand-

ing that territorial integrity is one of the most sacred conditions under which states work with each other. It is the basis of our cooperation and cannot be touched. And we have seen that the Russians don't care; they have no scruples about invading and they are willing to engage massively in the invasion of a country that is much smaller and weaker than them. It was a wake-up call, especially for my country, because after the war we believed we would live together in peace for eternity. That, unfortunately, did not turn out to be true. What we see now is Russia interfering heavily in every European country through cyber warfare. Just this morning we learned that there was a cyber attack on the electricity grid in Poland. It lasted only a couple of seconds and it would have succeeded if somebody had not intervened. Drones fly over airports all the time. So we

are, in a way, under attack: not direct attack, but cyber attack, and we therefore have to put ourselves in a new position to counter it. This war has now lasted longer than World War I. It shows that we have to be in a better position to defend ourselves — because who is to say the Russians will not attack Estonia, or some other country they feel should not exist?

Shubhajit Roy: You have come out in strong defence of NATO. But you have also seen the US commitment to the transatlantic partnership weaken, especially under President Trump. Last week, when he came for the NATO summit, he said he wouldn't have come if it were not Turkish President Recep Tayyip Erdoğan hosting it.

But the alliance stands, and what Trump

has, in a way, achieved is that every European country now spends more money on the military. And frankly, the Americans had a point. It was very comfortable for us — there was no threat, or no felt threat. Everybody was quite comfortable and said, let the Americans look after our security. But that was unfair, and everybody has understood that. In fact, previous US administrations made the same point — President Biden did, President Obama did — only more politely. The combination of Russia and Trump has led to a much more energetic resolve on this.

Sandeep Singh: Immigration is becoming a major electoral issue around the world. Do you see it as a temporary political shift, or a structural one?

The areas of Germany where there is the most suspicion of, or reluctance towards migration, are the parts where there is no migration. When people live in an environment where migration is normal, they don't feel anti-migration. In Germany's big cities, the AfD has a relatively small vote share, but when you go into the countryside, and especially into East Germany, where there is much less migration than in West Germany, it polls very high. It is psychological: there is a strong feeling of being overwhelmed, of being threatened in your way of life, cheated because people feel that migrants are undermining the social welfare system. So it is not rational. The question is what lies behind it and how do we heal it, work with it, cope with it. Life today is much more complicated than it was 20 years ago, and among these unhappy voters, this dissatisfaction with the present is also, in large part, nostalgia. People think life was much better in the old days. When I was growing up, a teacher in Germany could build his own house; that is no longer possible because everything has become so expensive. Globalisation is responsible for that. But one has to bring a certain pragmatism to it.

Shubhajit Roy: One of Germany's success stories has been skill development. Why has it not taken off in India? Is it about dignity of labour?

One difference between India and Germany is that craftsmanship is given dignity in Germany. I come from a very small town in Bavaria where the dignitaries were, basically, the craftsmen — not the notaries, the teachers or the doctors. The baker whose family has been there for 400 years, the butcher, the roofer: generations of good craftsmen selling good bread, sausage and beer. They were the ones running the town. They sat on the city council. That is not so much the case in India. Not yet.

Aakash Joshi: Remigration, which was once a fringe idea, is very real now. Once such ideas start gaining currency, the engagement is not always rational.

Remigration is a word Germans do not like to use; it is linked to the extreme Right. But what the current government has done is reduce illegal immigration. That shows the government has understood that people have this feeling of being overwhelmed by migration, and it has said: We will organise migration in such a way that those who are useful to our society come to Germany.

Aakash Joshi: Post Cold War, there was one model for emerging countries. Increasingly, there is a deeper engagement with the Chinese model and a questioning of the once championed liberal values.

The post-Cold War conviction that there is only one solution for the world — an open-minded liberal democracy in which everybody is free to find their own way — was not entirely correct. Besides China, there are other examples of countries that are performing well without being set up as Western liberal democracies, including Singapore and Vietnam. Rwanda is one of Africa's big success stories. So we have to accept that there are different ways of structuring societies — as long as they do not threaten the territorial integrity of their neighbours.

Jatin Anand: How do you evaluate India's diplomatic footprint, especially under PM Modi?

India is becoming more and more a factor that you cannot ignore. All of us — not only India but also Europe — find ourselves in an environment that is far more difficult than it used to be. The war in the Gulf has made things difficult and it has had an impact on all our countries. Foreign policy, therefore, is much more complicated than it used to be.

Jatin Anand: On the raising of your defence budget, largely because of Trump's demands: do you think there is actually space, and a need, for investment in conventional warfare amid AI and drone technology?

Yes. And it is not just because of Trump. The rearming of European nations is because of Russia. When you look at our investments, they are not only in conventional warfare but also in cyber warfare and other domains. We need a new generation of aircrafts. We have to invest heavily in modern war technology — drones and the like. And Ukraine will be one of the most interesting countries in the world from which to purchase non-conventional war material. Zelenskyy is travelling to Saudi Arabia and the UAE, and they are all interested in what Ukraine has to offer.

Kaushik Das Gupta: On the heat crisis in Europe: the cities were not built for such heat. What does that mean for green politics in Germany?

At the end of June, it was hotter in Berlin than here. The heatwave was linked to some 10,000 deaths in Europe. Air-conditioning is the first idea that comes to mind. But in southern European cities — in Spain, southern France and Italy — there are ways to make cities cooler without spending too much energy. Greenery is important, as are huge parasol-like structures for shade. We have to think about how to make our cities cooler and that architectural thinking is now reaching central and northern Europe too.

Raj Kamal Jha: In your public diplomacy you engaged with art, literature, culture. Now that you are leaving India, is there something that still puzzles you about the country? And as you head to Beijing, what is the big question on your mind about China?

India is a place that never stops surprising you. I will never say that I understand India. I understand parts of India, and I am emotional about them. But there are always situations where you think: how is that possible? I like that because India is always in flux. What is very striking is that in Europe there is a clear line between the rational and the irrational. In India, this line is blurred in a surprising and convincing way. As for China, everything one reads is about modern China — post-1945, Communist China, Mao Zedong. Unlike in India, where history, tradition and culture always play a big role, with China it is all about recent politics. But Chinese civilisation is thousands of years of art, culture and great inventions. I am very curious to discover the roots of China — roots that shaped today's China, some of which, like Buddhism, came from India.





THE HEALING TOUCH

DR ROMMEL TICKOO
SENIOR DIRECTOR, INTERNAL MEDICINE,
MAX HEALTHCARE



GETTY IMAGES

They simply have a much narrower role than marketing suggests

Why your Omega-3 supps may not be helping your heart

Research has found little benefit for most healthy adults

PICTURE THIS: You're standing in the supplement aisle, staring at a bottle of fish oil that promises to "support heart health." It seems like a sensible investment. You don't eat much fish, so popping a capsule every morning feels like an easy way to protect your heart. It's a belief shared by millions of people. In fact, about one in five adults over 60 takes fish oil supplements. The problem? The science behind those claims doesn't hold up nearly as well. For years, the idea that fish oil supplements prevent heart disease was largely built on observational studies showing that populations eating more fatty fish had lower rates of cardiovascular disease. But observational studies can only identify associations — they cannot prove cause and effect. People who regularly eat fish also tend to have healthier overall diets, exercise more and consume less processed meat, all of which independently reduce cardiovascular risk. When researchers finally put fish oil supplements to the test in large, randomised controlled trials — the gold standard of medical evidence — the results were disappointing. A landmark meta-analysis published in *JAMA Cardiology* found that Omega-3 supplements were not associated with reductions in heart attacks, strokes, coronary heart disease or major vascular events. In other words, the cardiovascular benefits seen in people who eat fish did not translate into swallowing an over-the-counter fish oil capsule.

One reason is that eating fish delivers far more than just Omega-3 fats. Fish provides high-quality protein, vitamins, minerals and replaces less healthy foods in the diet. Simply isolating Omega-3 into a capsule doesn't recreate those dietary benefits. Unlike prescription medications, dietary supplements are regulated as foods rather than drugs. Manufacturers are not required to demonstrate that their products actually prevent disease before selling them. An analysis of more than 2,800 fish oil supplement labels found that roughly 2,000 carried heart health claims, despite the lack of convincing evidence. Researchers have also found substantial variability in the actual Omega-3 content of commercially available supplements. Some products contain far less of these Omega-3 fats than consumers expect. Omega-3 fatty acids are chemically fragile and easily damaged by heat, oxygen and light during manufacturing and storage. Oxidised fish oil loses biological activity and may even promote inflammation rather than reduce it. Laboratory testing has shown that many commercially available fish oil products contain significant levels of oxidation.

So are Omega-3 supplements useless? Not at all. They simply have a much narrower role than marketing suggests. Evidence supports Omega-3 supplementation — of prescription strength and not OTCs — in certain situations. Prescription-strength EPA (icosapent ethyl) has been shown to reduce cardiovascular events in selected high-risk patients who have elevated triglycerides despite taking statins. Fish oil at doses above 2 grams per day can significantly lower triglyceride levels. Omega-3s are important for foetal brain and eye development during pregnancy. Regular dietary Omega-3 intake may support overall brain and eye health. These benefits, however, should not be confused with preventing heart disease in otherwise healthy people using low-dose, over-the-counter supplements. They aren't completely harmless either. Fish oil supplements can cause fishy burps, bloating, nausea and diarrhoea. Higher doses (typically above 4 grams per day) have also been associated with an increased risk of atrial fibrillation in some studies. Most over-the-counter capsules contain around 300–500 mg of Omega 3, well below prescription doses used to lower triglycerides.

Supplementation may be reasonable for people who eat less than one or two servings of fatty fish each week, those with elevated triglycerides (under medical supervision) and specific medical conditions. For everyone else, especially those with normal triglyceride levels, there is little evidence that routine OTC fish oil supplementation prevents cardiovascular disease.



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Reva Thakkar

AT 16, when Anushka Gaikwad stopped getting her periods, she did what most teenagers wouldn't — she visited the nearest clinic in her village of Cundaim in Goa. Instead of looking for an underlying hormonal problem, the doctor asked her to take a pregnancy test. Alarmed, Anushka persuaded her mother to seek another opinion in Panaji. A sonography finally revealed the real cause: Polycystic Ovarian Syndrome (PCOS), now referred to as Polyendocrine Metabolic Ovarian Syndrome (PMOS) to better reflect what doctors now understand about the condition.

Now 22, she says, "I realised later how little people, even healthcare providers, understood women's problems." Her experience is far from unique. PMOS is the most common endocrine disorder affecting women of reproductive age worldwide. But despite the name change, many still think of it as a disease of the ovaries. In reality, it is a metabolic disorder that affects nearly every system in the body — from insulin regulation and fertility to cardiovascular health, sleep, mood and mental wellbeing.

According to *The Lancet*, PMOS affects around one in eight women globally. In India, however, the burden is higher, with recent studies suggesting that one in five women may have the condition. "Almost one in five women below the age of 25 has some impact of PMOS, compared to the global average of around 8-13 per cent," says Dr Anshumala Shukla, head, Minimally Invasive Gynaecology, at Kokilaben Dhirubhai Ambani Hospital, Mumbai, who specialises in endometriosis, PMOS and is part of WHO efforts to draw up guidelines for the former. "Part of this is because India already has very high rates of diabetes and insulin resistance. Added to that are rapid urbanisation, easy access to processed food, sedentary lifestyles and rising stress levels," she says.

Why India is seeing early PMOS

The increase in diagnoses is partly because more women are seeking help for menstrual problems. "It's no longer taboo to see a gynaecologist," says Dr Meenakshi Ahuja, principal director, Obstetrics and Gynaecology, Fortis La Femme, Delhi. But awareness has not spread evenly. Women in rural areas often remain undiagnosed for years, while mothers — who are usually the first point of support for adolescent girls — have little understanding of the condition. India's underlying metabolic profile also makes women more vulnerable. A family history of diabetes and insulin resistance in-

One in five women now lives with PMOS

Why is it rising?

Once dismissed as an ovarian disorder, it is now understood to be a metabolic condition linked to insulin resistance, lifestyle and genetics

creases susceptibility to PMOS. While Dr Ahuja estimates India's overall prevalence at around 10 per cent, she says rates in cities — where sedentary routines and poor sleep are more common — may range between 25 and 40 per cent.

Dr Shukla believes the younger age at diagnosis is no coincidence. "Teenagers have higher screen times, eat unhealthy food and don't exercise regularly. Those with a genetic tendency start showing symptoms soon after they begin menstruating," she says. Then there is smoking, common among women in their 20s. This raises levels of free testosterone in the blood that manifests as acne, excess facial hair and hair loss. Smoking increases fasting insulin levels, worsening insulin resistance. This can lead to rapid weight gain and a higher risk of developing Type 2 diabetes. Smoking further triggers body inflammation and reduces your ovarian reserve (the number and quality of your eggs) by almost 20 per cent.

What are its symptoms?

Irregular periods remain the symptom that prompts most women to seek medical help but doctors say they are only one part of the picture. They look for irregular or delayed menstrual cycles caused by lack of ovulation, signs of excess male hormones such as facial hair or male-pattern hair loss, ultrasound evidence of enlarged ovaries with multiple follicles, and hormonal abnormalities in blood tests, including elevated Anti-Müllerian Hormone (AMH) levels, the last measuring the number of small, early-stage egg follicles in the ovaries. In women with PCOS, this follicle count is often very high.

The syndrome can also trigger weight gain, infertility, insulin resistance, sleep disturbances, anxiety and depression. For Nak-



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shatra Borgaonkar, the diagnosis came at 14. When her periods finally began, they came only once before disappearing for another five months. A scan confirmed PMOS. A decade later, she says the diagnosis itself was not the hardest part. "Nobody really explained what was happening to my body. I was simply told to stop eating junk food and eat small meals, an advice that pushed me towards years of unhealthy eating habits," she says. She describes herself as having spent years on the eating disorder spectrum, fuelled by emotional eating linked to hormonal changes she barely understood.

For Tahera, 23, it was only after beginning treatment that she realised excessive body hair and persistent acne were symptoms of PMOS. Once prescribed oral contraceptive pills, she noticed her skin clearing and body hair reducing. Dr Shukla says these emotional consequences are common. "High testosterone levels can lead to weight gain, which causes anxiety and depression," she says. Irregular periods themselves also contribute to anxiety. Stress worsens the condition by increasing cortisol, which increases insulin resistance, which in turn worsens PMOS. Poor sleep compounds the problem. "You can't sleep at 3 or 4 in the morning and wake up at 11. That disrupts the body's circadian rhythm," says Dr Shukla.

Not just about losing weight

For years, PMOS treatment has been reduced to a familiar refrain: lose weight. "But it has more to do with hormonal imbalance and insulin resistance," says Ritika Samad-dar, regional head, Clinical Nutrition and Dietetics, Max Healthcare. "If you don't address hormonal imbalance, any weight loss is likely to be temporary." That means focusing less on eating less and more on eating better. "The quality of food becomes more

important than simply cutting down the quantity. Protein and fibre are gradually increased over several weeks before patients are advised what to cut back on. Blanket denials backfire and leave young women feeling guilty rather than empowered," she says.

Dr Shukla agrees that lifestyle is the cornerstone of treatment. "You may have a genetic tendency for PMOS because of your family history," she says. "But if your lifestyle involves processed food, excess carbohydrates, little exercise and poor sleep, those genes begin to express themselves." Exercise, she adds, should not be viewed as a temporary fix for weight loss but as a long-term part of managing insulin resistance.

That is reflected in patient experiences too. Rachita Jain noticed her symptoms amplify after the birth of her second child. "I was completely occupied with the children and stopped taking care of myself and routines. Weight gain, bloating and swelling came first. Missed periods followed. I was diagnosed with PMOS at 34," she says. Exercise and diet changed that. Anushka noticed something similar. During college, her daily four-kilometre walk kept her menstrual cycles relatively regular. During holidays, when that routine disappeared, the irregularity often returned.

Medication also has a role but experts caution against the widespread misconception that everyone needs birth control pills. "They are recommended only for certain patients," says Dr Shukla. Women with highly irregular periods and significantly elevated male hormones may benefit from oral contraceptive pills, while others can be managed with medicines that improve insulin resistance or help regulate menstrual cycles.

The cost of living with PMOS

Managing PMOS is not just emotionally demanding; it can be financially draining. Anushka has undergone six ultrasounds since her diagnosis at 16, many of them because she moved cities. Whether so many scans are medically necessary is itself debatable. PMOS cysts are functional, explains Dr Ahuja. They form and disappear as part of the menstrual cycle and generally do not require repeated imaging once a diagnosis has been established. "The patient's symptoms and menstrual cycle are much better indicators of how treatment is progressing," she adds. A follow-up scan may be useful to confirm that ovarian size has normalised, but routine repeat imaging is avoidable.

What is needed are routine screening of adolescent girls, simple hormone and androgen tests and early diagnosis long before the condition worsens. "I wish we were informed and vocal about it," says Anushka.

• DECODING A STUDY

Air pollution may be quietly damaging your kidneys

A study of over 12,000 Indians found that long-term exposure to PM2.5 was linked to poor kidney function

Anonna Dutt

NOT JUST your lungs, high levels of air pollution may be damaging your kidneys too. Researchers found that increase in exposure to PM2.5 was linked to reduced filtration efficiency of the kidneys, according to a study that followed over 12,000 people for six to ten years. The findings were recently published in the journal *Kidney International Reports*. This decline was evident even when the researchers controlled other risk factors like diabetes and hypertension.

"This is part of an ongoing series of studies on the impact of air pollution on different diseases. So far, we have shown an association with higher air pollution levels and increased risk of hypertension, diabetes and dyslipidemia (high levels of blood fats like

cholesterol and triglycerides). Now, we have shown that pollutants impact kidney function as well. The association remained consistent even after controlling other factors, meaning the impact of air pollution on kidneys is not through the increases in diabetes and hypertension, but independently as well," says Dr V Mohan, one of the authors of the study and chairperson of Dr Mohan's Diabetes Specialties Centre, Chennai.

The participants were from Delhi and Chennai — two cities with very different levels of ambient air pollution. "The findings show that long-term exposure to fine air pollution may gradually reduce kidney function even in younger adults who do not already have chronic kidney disease (CKD)," adds Dr Mohan.

The PM2.5 levels in Delhi were much



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PM2.5 triggers systemic inflammation which damages cells and blood vessels

higher than Chennai during all three hospital visits by the participants during the study. The baseline PM2.5 level in Chennai stood at 33 mg/m3 and in Delhi 118 mg/m3. The levels stood at 37mg/m3 and 30 mg/m3 for Chennai during the subsequent visits

and 123 mg/m3 and 130 mg/m3 for Delhi.

The doctors measured the estimated glomerular filtration rates (eGFR) — used to check the health of the kidney — and found it to be 112 mL/min/1.73 m2 for Chennai residents and 106.4 for Delhi residents. While both values are considered to be normal, people in Delhi were already starting with lower reserves. The decline was also sharper. The study found that every 5 mg/m3 increase in annual average PM2.5 levels led to a decline of kidney function by 0.42 in Delhi residents and 0.32 in Chennai residents.

"There could be as much as four per cent decline in a year. Kidney function goes down with age but if you start with lower reserves, the damage would be quicker," says Dr Dora-ranj Prabhakaran, one of the authors and executive director of Centre for Chronic Disease Control. PM2.5 triggers systemic inflammation, oxidative stress that damages cells and blood vessels. The small particles directly enter the blood and then travel to the kidneys.

